

國立臺灣大學電機資訊學院生醫電子與資訊學研究所

碩士論文(初稿)

Graduate Institute of Biomedical Electronics and Bioinformatics

College of Electrical Engineering and Computer Science

National Taiwan University

Master's Thesis (Draft)

基於基因體基礎模型之總體基因體短讀序列分類學分  
類

Token-level Genomic Foundation Model Fine-tuning  
for Metagenomic Taxonomic Classification

楊明儒

Ming-Ju Yang

指導教授：劉子毓 博士

Advisor: Tzu-Yu Liu, Ph.D.

中華民國 115 年 7 月

July 2026





## 誌謝





# 基於基因體基礎模型之總體基因體短讀序列分類學分 類

研究生：楊明儒

指導教授：劉子毓 博士

國立臺灣大學 生醫電子與資訊學研究所

## 摘要

總體基因體學的分類學分類——即將每條定序短讀序列指派至正確的微生物分類單元——是微生物群落分析的核心計算挑戰。本論文系統性評估基因體基礎模型於總體基因體分類任務的適用性，並透過資料規模、預訓練、tokenization 等多重消融實驗，分析其主要瓶頸。我們提出一種基於 Nucleotide Transformer v2 (498M 參數) 的 Token 層級分類方法，結合低秩適應 (LoRA) 微調與注意力池化分類頭，保留完整的 Token 層級序列資訊。透過系統性的資料規模實驗 (500K 至 50M 條平衡讀序列)，我們發現在固定 NT-v2 架構下，訓練資料量是決定分類準確度的主要因素：在屬 (genus) 層級 120 類分類任務中，資料量由 500K 增至 5M 可提升準確度 7.76 個百分點 (55.29% 提升至 63.05%)，進一步擴至 50M 可達 67.07%，而架構與訓練技巧的調整最多僅帶來  $\pm 0.5$  個百分點的變化。反向互補測試時增強 (RC TTA) 在所有設定下穩定提供 0.1–1.5 個百分點的準確度改善。受控消融實驗證實預訓練 29 層骨幹網路比相同資料量下的單層隨機初始化模型高出 13.19 個百分點，驗證了預訓練表示在固定 tokenization 下的價值。然而，跨架構的 tokenization 消融實驗進一步顯示，在相同 50M 資料量下，採用重疊 13-mer tokenization 的 MetaTransformer 從頭訓練即可達到 87.42% 的屬層級準確度，較 NT-v2 + LoRA (67.07%) 高出 20.35 個百分點，顯示 tokenization 設計是比預訓練更具決定性的因素，而 NT-v2 的非重疊 6-mer tokenizer 是其主要結構性限制。在物種 (species) 層級 1,535 類任務上，per-genus 50M LoRA 分類器於 oracle genus

routing 下達到 27.8% Top-1，超越 sp\_v4 平面 oracle 上限 (27.4%)，驗證階層式架構方向的正確性。樣本層級評估顯示，在物種平衡的隨機分割合成樣本中，67% 的單讀序列準確度可轉化為相對豐度估計 Pearson 相關係數  $r = 0.993$  與豐度  $\geq 1\%$  之屬的 100% 偵測靈敏度，但此結果是在固定群落組成下的穩定性測試，於真實多樣性社群上的泛化仍待驗證。本研究為基因體基礎模型在總體基因體學分類上的應用提供了實證基礎與可擴展的實驗框架，並指出未來方向應聚焦於以重疊長  $k$ -mer tokenization 預訓練的微生物基礎模型。

**關鍵字：**

基因體基礎模型、總體基因體學、分類學分類、低秩適應、注意力池化

# Token-level Genomic Foundation Model Fine-tuning for Metagenomic Taxonomic Classification



Student: Ming-Ju Yang

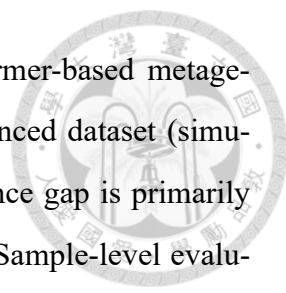
Advisor: Tzu-Yu Liu, Ph.D.

Graduate Institute of Biomedical Electronics and Bioinformatics  
National Taiwan University

## ABSTRACT

Metagenomic taxonomic classification—assigning each sequencing read to its source organism in the microbial taxonomy—is a fundamental computational challenge in microbiome analysis. This thesis proposes a token-level classification approach based on genomic foundation models, leveraging the pre-trained Nucleotide Transformer v2 (498M parameters) with Low-Rank Adaptation (LoRA) fine-tuning and an attention-pooling classification head that preserves the full token-level sequence information.

Through systematic data scaling experiments spanning 500K to 50M balanced training reads, we demonstrate that training data volume is the dominant factor determining classification accuracy: a  $10\times$  increase in training data yields a +7.76 percentage point improvement in genus-level accuracy (55.29% to 63.05% on 120 genera at 5M reads, scaling further to 67.07% at 50M reads), while architectural and training technique ablations produce at most  $\pm 0.5$  percentage points of variation. Reverse-complement test-time augmentation consistently improves accuracy by +0.1 to +1.5 percentage points across all experimental settings. Balanced subsampling across species reduces the number of unclassifiable taxa ( $F1=0$  classes) from 9 to 0 at the 50M scale and improves macro F1 by +5.08 percentage points over the 5M baseline.



Comparison with MetaTransformer, a state-of-the-art Transformer-based metagenomic classifier achieving 98.3% genus recall on a large-scale balanced dataset (simulated from 2,505 HGR-UMGS genomes), reveals that the performance gap is primarily attributable to training data volume rather than model architecture. Sample-level evaluation over synthetic 50,000-read communities demonstrates that the 67% per-read accuracy translates to Pearson  $r = 0.993$  for relative abundance estimation and 100% genus detection sensitivity for taxa at  $\geq 1\%$  relative abundance. This work provides empirical evidence for the data scaling requirements of genomic foundation models in metagenomic classification and establishes a scalable experimental framework for further investigation.

**Keywords:**

Genomic Foundation Model, Metagenomics, Taxonomic Classification, LoRA, Attention Pooling

# CONTENTS



誌謝

摘要

ABSTRACT

CONTENTS

LIST OF FIGURES

LIST OF TABLES

List of Symbols and Abbreviations

Chapter 1 Introduction

|     |                                     |   |
|-----|-------------------------------------|---|
| 1.1 | Background and Motivation . . . . . | 1 |
| 1.2 | Problem Statement . . . . .         | 3 |
| 1.3 | Contributions . . . . .             | 4 |
| 1.4 | Thesis Organization . . . . .       | 5 |

Chapter 2 Background and Related Work

|       |                                                         |    |
|-------|---------------------------------------------------------|----|
| 2.1   | Metagenomic Taxonomic Classification . . . . .          | 7  |
| 2.1.1 | Alignment-Based Methods . . . . .                       | 7  |
| 2.1.2 | $k$ -mer-Based Methods . . . . .                        | 8  |
| 2.1.3 | Deep Learning Methods . . . . .                         | 8  |
| 2.2   | Genomic Foundation Models . . . . .                     | 10 |
| 2.2.1 | Nucleotide Transformer . . . . .                        | 10 |
| 2.2.2 | Other Genomic Foundation Models . . . . .               | 11 |
| 2.2.3 | GFM for Metagenomic Classification . . . . .            | 12 |
| 2.3   | Parameter-Efficient Fine-Tuning . . . . .               | 13 |
| 2.3.1 | Low-Rank Adaptation (LoRA) . . . . .                    | 13 |
| 2.3.2 | Other PEFT Methods . . . . .                            | 14 |
| 2.4   | Attention Mechanisms for Sequence Aggregation . . . . . | 15 |
| 2.4.1 | Mean Pooling . . . . .                                  | 15 |
| 2.4.2 | Attention Pooling . . . . .                             | 15 |
| 2.4.3 | Multi-Head Attention Pooling . . . . .                  | 16 |



|                  |                                                     |           |
|------------------|-----------------------------------------------------|-----------|
| 2.5              | Data Augmentation for DNA Sequences . . . . .       | 16        |
| 2.5.1            | Reverse-Complement Symmetry in DNA . . . . .        | 16        |
| 2.5.2            | Strategies for Exploiting RC Symmetry . . . . .     | 17        |
| 2.6              | Handling Class Imbalance . . . . .                  | 19        |
| 2.6.1            | Loss-Level Approaches . . . . .                     | 19        |
| 2.6.2            | Sampling-Level Approaches . . . . .                 | 20        |
| 2.7              | Summary . . . . .                                   | 20        |
| <b>Chapter 3</b> | <b>Methodology</b>                                  | <b>21</b> |
| 3.1              | System Overview . . . . .                           | 21        |
| 3.2              | Backbone: Nucleotide Transformer v2 . . . . .       | 22        |
| 3.2.1            | Tokenization . . . . .                              | 23        |
| 3.2.2            | Encoder Architecture . . . . .                      | 24        |
| 3.2.3            | LoRA Adaptation . . . . .                           | 25        |
| 3.2.4            | Ablation: Shallow Transformer Backbone . . . . .    | 25        |
| 3.3              | Classification Head: Attention Pooling . . . . .    | 27        |
| 3.3.1            | Architecture . . . . .                              | 27        |
| 3.3.2            | Alternative Heads . . . . .                         | 28        |
| 3.4              | Training Strategy . . . . .                         | 28        |
| 3.4.1            | Two-Phase Training . . . . .                        | 28        |
| 3.4.2            | Optimization . . . . .                              | 29        |
| 3.4.3            | Transfer Learning Across Experiments . . . . .      | 30        |
| 3.5              | Data Pipeline . . . . .                             | 30        |
| 3.5.1            | Training Data Generation . . . . .                  | 30        |
| 3.5.2            | Balanced Subsampling . . . . .                      | 31        |
| 3.5.3            | Reverse-Complement Augmentation . . . . .           | 31        |
| 3.5.4            | Data Splitting . . . . .                            | 32        |
| 3.6              | Evaluation Methodology . . . . .                    | 32        |
| 3.6.1            | Metrics . . . . .                                   | 32        |
| 3.6.2            | Reverse-Complement Test-Time Augmentation . . . . . | 33        |
| 3.6.3            | Species-Level Evaluation Modes . . . . .            | 33        |
| 3.7              | Computational Considerations . . . . .              | 34        |
| 3.7.1            | Hardware . . . . .                                  | 34        |
| 3.7.2            | Memory Optimization . . . . .                       | 34        |
| 3.7.3            | Resource Monitoring . . . . .                       | 34        |

|     |                   |    |
|-----|-------------------|----|
| 3.8 | Summary . . . . . | 35 |
|-----|-------------------|----|

## Chapter 4 Experiments and Results

|         |                                                              |    |
|---------|--------------------------------------------------------------|----|
| 4.1     | Experimental Setup . . . . .                                 | 37 |
| 4.1.1   | Dataset . . . . .                                            | 37 |
| 4.1.2   | Evaluation Protocol . . . . .                                | 38 |
| 4.1.3   | Hardware and Training Infrastructure . . . . .               | 38 |
| 4.2     | Phase 1: Frozen Embedding Baselines . . . . .                | 38 |
| 4.2.1   | Setup . . . . .                                              | 38 |
| 4.2.2   | Results . . . . .                                            | 39 |
| 4.3     | Phase 2: Token-Level Classification with LoRA . . . . .      | 39 |
| 4.3.1   | Architecture Validation (v1–v3, 500K Dataset) . . . . .      | 39 |
| 4.3.2   | Training Dynamics and Overfitting . . . . .                  | 40 |
| 4.4     | Advanced Training Techniques (v4–v7, 500K Dataset) . . . . . | 42 |
| 4.5     | Data Scaling Experiments . . . . .                           | 43 |
| 4.5.1   | Experiment v8: 5M Balanced Reads . . . . .                   | 43 |
| 4.5.1.1 | Setup . . . . .                                              | 43 |
| 4.5.1.2 | Results . . . . .                                            | 43 |
| 4.5.1.3 | Training Dynamics . . . . .                                  | 44 |
| 4.5.2   | Experiment v9: 50M Balanced Reads . . . . .                  | 45 |
| 4.5.2.1 | Results . . . . .                                            | 45 |
| 4.5.3   | Data Scaling Analysis . . . . .                              | 47 |
| 4.6     | Species-Level Classification . . . . .                       | 49 |
| 4.6.1   | Direct Classification . . . . .                              | 49 |
| 4.6.2   | Oracle-Genus Evaluation . . . . .                            | 49 |
| 4.6.3   | Effect of Class Imbalance Mitigation . . . . .               | 49 |
| 4.6.4   | Top- $k$ Genus Routing . . . . .                             | 50 |
| 4.7     | RC Test-Time Augmentation Analysis . . . . .                 | 50 |
| 4.8     | Computational Resource Analysis . . . . .                    | 52 |
| 4.8.1   | Observed Resource Usage . . . . .                            | 52 |
| 4.8.2   | Bottleneck Analysis . . . . .                                | 52 |
| 4.8.3   | Scaling Projections . . . . .                                | 53 |
| 4.9     | Comparison with MetaTransformer . . . . .                    | 54 |
| 4.10    | Backbone Ablation: Shallow Transformer . . . . .             | 55 |
| 4.10.1  | Experimental Design . . . . .                                | 55 |

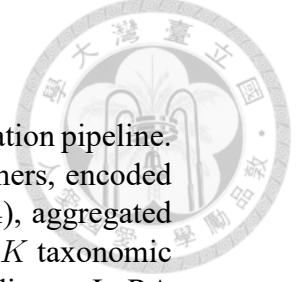
|                  |                                                                 |           |
|------------------|-----------------------------------------------------------------|-----------|
| 4.10.2           | Expected Outcomes . . . . .                                     | 56        |
| 4.10.3           | Results . . . . .                                               | 56        |
| 4.11             | Tokenization Ablation: MetaTransformer at 50M Scale . . . . .   | 58        |
| 4.12             | Species-Level Classification at Scale . . . . .                 | 63        |
| 4.12.1           | Experimental Design . . . . .                                   | 64        |
| 4.12.2           | Final Results (sp_v4, Epoch 30) . . . . .                       | 64        |
| 4.12.3           | Hierarchical Classification Pipeline . . . . .                  | 66        |
| 4.12.4           | Hierarchical Evaluation on an Independent Test Set . . . . .    | 67        |
| 4.12.5           | Per-Genus Model Analysis . . . . .                              | 68        |
| 4.13             | Sample-Level Application Evaluation . . . . .                   | 69        |
| 4.13.1           | Experimental Setup . . . . .                                    | 69        |
| 4.13.2           | Relative Abundance Estimation . . . . .                         | 70        |
| 4.13.3           | Genus Detection (Binary Presence/Absence) . . . . .             | 73        |
| 4.13.4           | Summary . . . . .                                               | 75        |
| 4.13.5           | Comparison with MetaTransformer . . . . .                       | 76        |
| 4.14             | Summary of Findings . . . . .                                   | 77        |
| <b>Chapter 5</b> | <b>Conclusion and Future Work</b>                               | <b>79</b> |
| 5.1              | Summary of Contributions . . . . .                              | 79        |
| 5.2              | Limitations . . . . .                                           | 81        |
| 5.3              | Future Work . . . . .                                           | 82        |
| 5.3.1            | Domain-Specific Pre-trained Foundation Models . . . . .         | 82        |
| 5.3.2            | Full-Scale Data Training . . . . .                              | 83        |
| 5.3.3            | Species-Level Scaling and Hierarchical Classification . . . . . | 83        |
| 5.3.4            | Tokenization Study . . . . .                                    | 84        |
| 5.3.5            | Architectural Improvements . . . . .                            | 84        |
| 5.3.6            | Real-World Validation . . . . .                                 | 85        |
| 5.3.7            | Efficiency Optimization . . . . .                               | 85        |
| 5.3.8            | Pre-training Strategy . . . . .                                 | 86        |
| 5.4              | Concluding Remarks . . . . .                                    | 86        |
|                  | <b>References</b>                                               | <b>89</b> |
|                  | <b>Appendix A Experiment Configuration Details</b>              | <b>95</b> |
| A.1              | v8 Genus Configuration (5M Balanced) . . . . .                  | 95        |
| A.2              | v9 Genus Configuration (50M Balanced) . . . . .                 | 96        |

|     |                                    |    |
|-----|------------------------------------|----|
| A.3 | SLURM Job Script Example . . . . . | 97 |
|-----|------------------------------------|----|

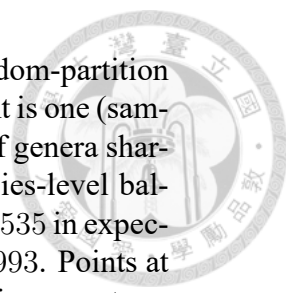




# LIST OF FIGURES



- Figure 3.1 Overview of the proposed token-level GFM classification pipeline. A 150 bp DNA read is tokenized into non-overlapping 6-mers, encoded by the LoRA-adapted NT-v2 backbone (29 layers,  $d=1024$ ), aggregated via multi-head attention pooling, and classified into one of  $K$  taxonomic classes (120 genera or 1,535 species). Orange badge indicates LoRA adapter modules; dashed boxes delineate the backbone and classification head parameter groups. . . . . 22
- Figure 4.1 Training dynamics for v3 (500K, left), v8 (5M, centre), and v9 (50M, right). Dashed lines show training accuracy; solid lines show validation accuracy. The shaded region highlights the train-validation gap. Dotted vertical lines mark learning-rate resets caused by SLURM timeout and job resumption; v9 experienced two such resets (epochs 7 and 11) before the resume logic was corrected. Overfitting decreases systematically as data volume increases. . . . . 41
- Figure 4.2 Full validation accuracy curves for v9 (genus, left) and sp\_v4 (species, right) over all completed epochs. Red dotted vertical lines mark epochs where a SLURM job restart incorrectly reset the cosine LR scheduler, causing a temporary accuracy dip. The green shaded region indicates epochs trained after the resume bug fix was applied, during which the LR schedule correctly continues its cosine decay. . . . . 47
- Figure 4.3 Genus RC TTA accuracy versus training data volume (log scale). Blue circles mark our results at three data scales; the red star marks MetaTransformer’s reported genus recall. The dashed line is a log-linear fit to our three data points. Per- $10\times$  accuracy gains diminish from  $+7.76$  pp (500K $\rightarrow$ 5M) to  $+4.02$  pp (5M $\rightarrow$ 50M). The MetaTransformer star marks reported genus recall (98.3%); its x-position is based on training throughput (300K steps  $\times$  batch 2,048) and is not directly comparable to our dataset-size axis. . . . . 48
- Figure 4.4 RC TTA benefit across experiments. Top: forward-only vs. RC TTA accuracy (blue: NT-v2 backbone; green: DNABERT family; right of second dashed line: random init). Bottom: absolute gain in percentage points. NT-v2-based models gain  $+0.77$ – $+1.54$  pp consistently. DNABERT gains only  $+0.58$  pp, likely because its overlapping 6-mer tokenization is inherently more RC-symmetric than NT-v2’s non-overlapping 6-mer. DNABERT-2 (BPE tokenization) gains  $+1.53$  pp, similar to early NT-v2 experiments. The randomly initialized v11 gains only  $+0.08$  pp, confirming that the benefit primarily corrects pre-training-induced strand biases. . . . . 52
- Figure 4.5 Backbone ablation: pre-trained NT-v2 + LoRA (v9, green) vs. single-layer randomly initialized Transformer (v11, purple), both trained on 50M balanced reads. Numbers above bars indicate the absolute gain of v9 over v11. The pre-trained backbone consistently outperforms the shallow variant across all metrics, with the largest gap in F1 (macro) ( $+19.0$  pp) and Balanced Accuracy ( $+17.6$  pp), reflecting v9’s superior discrimination of rare genera. . . . . 58



- Figure 4.6 True vs. predicted relative abundance across 100 random-partition samples (all 12,000 (sample, genus) pairs shown). Each point is one (sample, genus) pair. The visible clusters correspond to groups of genera sharing the same species count  $n_g$ : because the dataset is species-level balanced, every genus with  $n_g$  species has true abundance  $n_g/1,535$  in expectation, collapsing to the same  $x$ -coordinate. Pearson  $r = 0.993$ . Points at larger  $x$ -values drift slightly above the identity line, reflecting a systematic overestimation of high-abundance genera attributable to classification bias (Section 4.13.2). . . . . 72
- Figure 4.7 Sample-level genus detection sensitivity by true relative abundance (detection threshold:  $\geq 1$  predicted read), aggregated over 200 sparse-community samples. Annotation shows the number of (sample, genus) instances in each bucket. . . . . 73
- Figure 4.8 ROC curve for genus presence/absence detection. Ground truth is binary (from sparse-community construction); detection score is predicted relative abundance. Aggregated over 200 sparse-community samples (50K reads, 60/120 genera present each). Key operating points at fixed specificity targets are marked. . . . . 74

# LIST OF TABLES



|            |                                                                                                                                                                                                                                                         |    |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Table 3.1  | Token counts for a 150 bp read under different $k$ -mer tokenization settings. . . . .                                                                                                                                                                  | 24 |
| Table 3.2  | LoRA hyperparameters. . . . .                                                                                                                                                                                                                           | 25 |
| Table 3.3  | Comparison of backbone configurations. . . . .                                                                                                                                                                                                          | 26 |
| Table 3.4  | Training optimization hyperparameters. . . . .                                                                                                                                                                                                          | 29 |
| Table 4.1  | Frozen embedding baselines for genus classification (120 classes, 500K dataset). . . . .                                                                                                                                                                | 39 |
| Table 4.2  | Hyperparameter configurations for initial architecture experiments (500K dataset). Bold values indicate the setting that changed relative to v1. . . . .                                                                                                | 40 |
| Table 4.3  | Results for architecture experiments (500K dataset, genus classification). . . . .                                                                                                                                                                      | 40 |
| Table 4.4  | Training dynamics for v3 (500K dataset). The train-validation gap grows steadily, reaching 7% at the best epoch and 9% by epoch 35. . . . .                                                                                                             | 41 |
| Table 4.5  | Advanced training technique experiments (500K dataset, genus classification). RC TTA results are shown where available. . . . .                                                                                                                         | 42 |
| Table 4.6  | Genus classification results across data scales: v4 (500K), v8 (5M balanced), and v9 (50M balanced). . . . .                                                                                                                                            | 44 |
| Table 4.7  | Training dynamics for v8 (5M balanced, 30 epochs). The gap between training and validation accuracy remains small throughout, in sharp contrast to the 500K experiments. . . . .                                                                        | 45 |
| Table 4.8  | Configuration comparison between v8 and v9. . . . .                                                                                                                                                                                                     | 45 |
| Table 4.9  | Comparison between v8 (5M) and v9 (50M) for genus classification (120 classes). . . . .                                                                                                                                                                 | 46 |
| Table 4.10 | Data scaling analysis: genus classification performance across data volumes. . . . .                                                                                                                                                                    | 47 |
| Table 4.11 | Species classification results (500K dataset, 1,535 classes). . . . .                                                                                                                                                                                   | 49 |
| Table 4.12 | Effect of RC TTA across experiments. RC TTA consistently improves accuracy at the cost of $2\times$ inference time. . . . .                                                                                                                             | 50 |
| Table 4.13 | Computational resource usage across experiments. . . . .                                                                                                                                                                                                | 53 |
| Table 4.14 | Resource bottleneck assessment on TWCC H100 nodes. . . . .                                                                                                                                                                                              | 53 |
| Table 4.15 | Comparison between our approach and MetaTransformer [8]. . . . .                                                                                                                                                                                        | 54 |
| Table 4.16 | Controlled ablation: NT-v2 backbone (v9) vs. shallow Transformer (v11). All other components are held constant. . . . .                                                                                                                                 | 55 |
| Table 4.17 | Backbone ablation results: pre-trained NT-v2 (v9) vs. shallow Transformer (v11), both trained on 50M balanced reads. . . . .                                                                                                                            | 57 |
| Table 4.18 | Tokenization ablation (genus-level, 120 classes): NT-v2 + LoRA vs. MetaTransformer under six tokenization configurations. All MetaTransformer models are trained from scratch on 50M balanced reads. RC TTA is unavailable for MetaTransformer. . . . . | 59 |
| Table 4.19 | Tokenization ablation (species-level, 1,535 classes): MetaTransformer under six tokenization configurations, trained on the same 50M balanced reads. . . . .                                                                                            | 59 |
| Table 4.20 | Pre-trained GFM backbone comparison at 5M balanced reads (genus-level, 120 classes). All models use LoRA $r = 16$ fine-tuning and the same attention-pooling head. RC TTA is applied to all models. . . . .                                             | 61 |
| Table 4.21 | Species classification scaling: 500K (sp_v1–v3) vs. 50M (sp_v4). . . . .                                                                                                                                                                                | 64 |



|            |                                                                                                                                                                                                                                                                                                                                |    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Table 4.22 | sp_v4 training progress (species classification, 1,535 classes, 50M reads). . . . .                                                                                                                                                                                                                                            | 64 |
| Table 4.23 | sp_v4 final forward-only evaluation on the 5M held-out set (epoch 30 checkpoint). . . . .                                                                                                                                                                                                                                      | 65 |
| Table 4.24 | Complete hierarchical species evaluation on the independent 100K-read test set. sp_v4 (ep30, 50M) denotes the flat species model with four routing modes. Per-genus (50M) denotes 81 dedicated classifiers, each trained on a 50M-read subset with a frozen NT-v2 backbone. Genus routing uses v9 (66.1% on this set). . . . . | 67 |
| Table 4.25 | Relative abundance estimation accuracy (100 random-partition samples, 50K reads each, all 120 genera present). . . . .                                                                                                                                                                                                         | 70 |
| Table 4.26 | Detection sensitivity by true relative abundance (detection threshold: $\geq 1$ predicted read). Genera truly absent from the sample are excluded from sensitivity computation. . . . .                                                                                                                                        | 73 |
| Table 4.27 | Sample-level evaluation across three models. Same evaluation protocol: 100 random-partition samples and 200 sparse-community samples, each 50K reads, 60/120 genera present per sparse sample. . . . .                                                                                                                         | 76 |

# List of Symbols and Abbreviations



## Abbreviations

|        |                                                  |
|--------|--------------------------------------------------|
| AMP    | Automatic Mixed Precision                        |
| ART    | Artificial Read Technology (read simulator)      |
| BPE    | Byte Pair Encoding                               |
| DNA    | Deoxyribonucleic Acid                            |
| ESM    | Evolutionary Scale Modeling                      |
| FFN    | Feed-Forward Network                             |
| GFM    | Genomic Foundation Model                         |
| GPU    | Graphics Processing Unit                         |
| HGR    | Human Gastrointestinal Reference                 |
| LA     | Logit Adjustment                                 |
| LCA    | Lowest Common Ancestor                           |
| LoRA   | Low-Rank Adaptation                              |
| LR     | Learning Rate                                    |
| MLM    | Masked Language Modeling                         |
| MLP    | Multi-Layer Perceptron                           |
| NT     | Nucleotide Transformer                           |
| PEFT   | Parameter-Efficient Fine-Tuning                  |
| RC     | Reverse Complement                               |
| RC TTA | Reverse-Complement Test-Time Augmentation        |
| RoPE   | Rotary Position Embedding                        |
| SLURM  | Simple Linux Utility for Resource Management     |
| TTA    | Test-Time Augmentation                           |
| UMGS   | Unified Human Gastrointestinal Genome collection |
| VRAM   | Video Random Access Memory                       |

## Symbols

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| $r$             | LoRA rank                                             |
| $\alpha$        | LoRA scaling factor                                   |
| $K$             | Number of classes                                     |
| $T$             | Sequence length (number of tokens)                    |
| $d$             | Hidden dimension of the backbone model                |
| $M$             | Number of learnable query tokens in attention pooling |
| $W_Q, W_K, W_V$ | Query, key, and value projection matrices             |
| $A, B$          | LoRA low-rank decomposition matrices                  |
| $\tau$          | Temperature parameter for logit adjustment            |
| $\pi_k$         | Prior probability of class $k$                        |
| $f_\theta$      | Classifier parameterized by $\theta$                  |
| $\bar{r}$       | Reverse complement of read $r$                        |



# Chapter 1

## Introduction



### 1.1 Background and Motivation

Metagenomics, the culture-independent study of microbial communities through direct sequencing of environmental DNA, has transformed our understanding of microbial ecology, human health, and environmental science [1]. Modern metagenomic sequencing platforms, particularly Illumina short-read technologies, generate hundreds of millions of reads per sample, each typically 100–300 base pairs (bp) in length. A fundamental computational challenge in metagenomic analysis is *taxonomic classification*: assigning each sequencing read to the correct taxon in the microbial taxonomy, such as genus or species [2].

Accurate taxonomic classification underpins virtually all downstream metagenomic analyses, including community composition profiling, pathogen detection, antibiotic resistance gene tracking, and comparative microbiome studies [3]. However, the classification task is inherently difficult due to several factors:

- **Short read length:** Individual reads of 150 bp carry limited phylogenetic signal, especially when distinguishing closely related taxa.
- **High taxonomic diversity:** A typical metagenomic sample may contain reads from hundreds to thousands of species spanning dozens of genera.
- **Class imbalance:** Community compositions follow highly skewed abundance distributions, with a few dominant taxa and many rare ones.

- **Sequence similarity:** Closely related taxa share substantial genomic homology, making fine-grained discrimination (e.g., species-level) particularly challenging.



Traditional approaches to metagenomic classification fall broadly into two categories. *Alignment-based methods* such as BLAST [4] and DIAMOND [5] align reads against reference databases, achieving high precision but at considerable computational cost. *k-mer-based methods* such as Kraken2 [6] and Centrifuge [7] index reference genomes using *k*-mer hash tables, enabling ultra-fast classification at the expense of sensitivity. More recently, deep learning approaches have emerged, including MetaTransformer [8], which applies a Transformer architecture to learned *k*-mer embeddings for metagenomic read classification.

Concurrently, the field of genomics has witnessed the development of *genomic foundation models* (GFMs)—large-scale pre-trained models that learn general-purpose representations of DNA sequences from billions of nucleotides [9]. Models such as the Nucleotide Transformer [9], DNABERT [10], DNABERT-2 [11], and Evo [12] have demonstrated strong performance across diverse genomic tasks, including promoter prediction, splice site identification, and variant effect prediction. These models offer a compelling alternative to task-specific feature engineering: by pre-training on vast genomic corpora, they capture rich sequence semantics that can be transferred to downstream tasks through fine-tuning.

Despite their promise, the application of GFMs to metagenomic taxonomic classification remains underexplored. A naïve approach—extracting mean-pooled embeddings from a frozen pre-trained model and training a linear classifier—discards the positional and local information encoded in individual token representations. This thesis investigates

whether a *token-level* approach, which preserves the full sequence of token embeddings and employs parameter-efficient fine-tuning, can substantially improve metagenomic taxonomic classification.



## 1.2 Problem Statement

Given a set of metagenomic short reads  $\{r_1, r_2, \dots, r_N\}$  generated by sequencing a microbial community, the taxonomic classification problem is to assign each read  $r_i$  to its correct taxon  $y_i \in \{1, 2, \dots, K\}$ , where  $K$  is the number of classes at a given taxonomic level (e.g., genus or species).

This thesis formulates the problem as a supervised multi-class classification task. We assume access to a reference genome database from which simulated reads with known taxonomic labels can be generated for training. The challenge is to learn a classifier  $f_\theta : \mathcal{R} \rightarrow \{1, \dots, K\}$  that generalizes well from simulated training data to accurately classify novel reads.

The specific objectives of this thesis are:

1. **Token-level representation:** Investigate whether preserving the full sequence of token embeddings from a genomic foundation model, rather than using mean-pooled summaries, improves taxonomic classification accuracy.
2. **Parameter-efficient fine-tuning:** Evaluate the effectiveness of LoRA (Low-Rank Adaptation) [13] for adapting a 498M-parameter pre-trained model to the taxonomic classification task with minimal trainable parameters.
3. **Data scaling:** Quantify the relationship between training data volume and classifi-

cation performance, and determine the data requirements for achieving competitive accuracy.

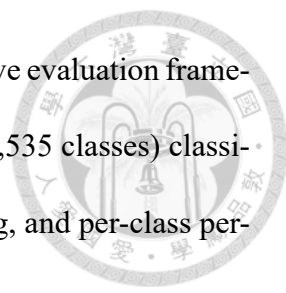


4. **Comprehensive evaluation:** Assess performance across multiple metrics (micro accuracy, macro F1, top- $k$  accuracy) and at both genus and species taxonomic levels.

## 1.3 Contributions

The main contributions of this thesis are as follows:

1. **A token-level GFM classification framework:** We propose a classification pipeline that feeds the complete token-level output of a pre-trained Nucleotide Transformer v2 [9] into a learnable attention-pooling classification head, fine-tuned with LoRA. This approach preserves rich positional and contextual information that is lost in mean-pooling approaches.
2. **Systematic data scaling analysis:** Through extensive experiments spanning 500K to 50M training reads, we demonstrate that training data volume is the dominant factor determining classification accuracy. Increasing training data from 500K to 5M balanced reads yields a +7.8 percentage point improvement in genus-level accuracy (55.3%  $\rightarrow$  63.1%), while architectural modifications alone yield at most  $\pm 0.5$  percentage points.
3. **Reverse-complement test-time augmentation:** We demonstrate that averaging logits from forward and reverse-complement passes at inference time consistently improves accuracy by +0.1 to +1.5 percentage points at no additional training cost, with larger gains for pre-trained models and smaller gains for randomly initialized architectures.

- 
4. **Multi-level taxonomic evaluation:** We provide a comprehensive evaluation framework supporting genus-level (120 classes) and species-level (1,535 classes) classification, including oracle-genus evaluation, top- $k$  genus routing, and per-class performance analysis.
  5. **Controlled backbone ablation:** We design a controlled experiment comparing the pre-trained 29-layer NT-v2 backbone against a single-layer Transformer trained from scratch (MetaTransformer-style), using the same tokenizer and classification head. This isolates the contribution of pre-trained representations from model capacity and tokenization effects.
  6. **Computational resource assessment:** We provide a detailed analysis of GPU memory, compute utilization, and training time on NVIDIA H100 hardware, offering practical guidance for scaling GFM-based classifiers to production metagenomic datasets.

## 1.4 Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2** reviews the background and related work, covering metagenomic classification methods, genomic foundation models, parameter-efficient fine-tuning, and attention mechanisms.
- **Chapter 3** presents the proposed methodology in detail, including the model architecture, training strategy, data pipeline, and evaluation framework.
- **Chapter 4** describes the experimental setup and presents comprehensive results,



including ablation studies, data scaling analysis, species-level evaluation, and computational resource assessment.

- **Chapter 5** summarizes the findings, discusses limitations, and outlines directions for future work.

# Chapter 2

## Background and Related Work



This chapter provides the necessary background for understanding the methods proposed in this thesis. We review existing approaches to metagenomic taxonomic classification, introduce genomic foundation models, describe parameter-efficient fine-tuning techniques, and discuss attention-based sequence aggregation mechanisms.

### 2.1 Metagenomic Taxonomic Classification

Metagenomic taxonomic classification aims to assign each sequencing read from an environmental sample to its source organism’s position in the phylogenetic tree. We review three major paradigms for this task.

#### 2.1.1 Alignment-Based Methods

Alignment-based methods classify reads by aligning them to reference genome databases. BLAST [4] performs local sequence alignment using heuristic seed-and-extend strategies, achieving high sensitivity but at significant computational cost (hours to days for typical metagenomic datasets). DIAMOND [5] accelerates protein-level alignments through double indexing and optimized scoring, achieving up to  $20,000\times$  speedup over BLAST while maintaining comparable sensitivity. MegaBLAST [14] optimizes nucleotide-level alignment for closely related sequences.

The primary limitation of alignment-based methods is their computational cost, which scales poorly with the size of modern metagenomic datasets containing hundreds of mil-

lions of reads.



### 2.1.2 $k$ -mer-Based Methods

$k$ -mer-based methods represent sequences as sets or distributions of fixed-length sub-sequences ( $k$ -mers) and use hash-table lookups for rapid classification.

**Kraken2** [6] constructs a database mapping each  $k$ -mer ( $k = 35$ ) to its lowest common ancestor (LCA) in the taxonomic tree. Classification proceeds by querying each read's  $k$ -mers against this database and assigning the read to the taxon that covers the most  $k$ -mers. Kraken2 achieves classification speeds exceeding  $10^6$  reads per minute, making it one of the most widely adopted tools in metagenomic analysis.

**Centrifuge** [7] uses a compressed FM-index to perform rapid exact matching of reads against reference genomes, achieving lower memory requirements than Kraken2 while maintaining competitive accuracy.

While  $k$ -mer methods excel in speed, they rely on exact  $k$ -mer matches and thus struggle with sequences that diverge from the reference database. They also lack the ability to capture higher-order sequence patterns beyond individual  $k$ -mers.

### 2.1.3 Deep Learning Methods

Deep learning methods learn discriminative representations directly from sequence data, potentially capturing complex patterns that elude  $k$ -mer matching.

**DeepMicrobes** [15] applies a bidirectional LSTM with attention to one-hot encoded nucleotide sequences, achieving competitive performance on species-level classification

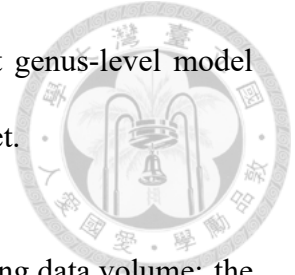
but requiring substantial training data and compute.

**BERTax** [16] adapts the BERT architecture to DNA taxonomic classification at superkingdom, phylum, and genus levels. Rather than relying on sequence similarity to a reference database, BERTax learns contextual  $k$ -mer representations and demonstrates generalisation to novel organisms with no close relatives in the training set. This work establishes the viability of transformer-based language models for read-level taxonomic assignment, motivating the use of larger pre-trained GFM in this thesis.

**MetaTransformer** [8] is the most directly relevant prior work to this thesis. It employs a custom Transformer architecture that operates on learned 12-mer embeddings. Key design choices include:

- **12-mer tokenization:** Input reads are segmented into non-overlapping 12-mers, each mapped to a learnable embedding vector. The choice of  $k = 12$  balances vocabulary size ( $4^{12} \approx 16.7\text{M}$  possible 12-mers, though only  $\sim 1.4\text{M}$  appear in training data) against sequence granularity.
- **Transformer encoder:** A Transformer encoder (the best-performing configuration uses  $d_{\text{model}} = 64$  with a single encoder block for  $k = 13$ ; the authors also evaluate up to 4 blocks and  $d_{\text{model}} = 128$ ) processes the sequence of  $k$ -mer embeddings.
- **Mean pooling:** Token representations are mean-pooled into a fixed-length vector for classification.
- **Large-scale training:** The model is trained on balanced reads simulated from 2,505 HGR-UMGS genomes using the ART Illumina simulator with a coverage factor of 3 (the exact total number of reads is not reported in the paper; we estimate  $\sim 614\text{M}$

based on 300K training steps  $\times$  batch size 2,048). The best genus-level model achieves 98.9% precision and 98.3% recall on the validation set.



A key insight from MetaTransformer is the critical role of training data volume: the authors demonstrate that performance scales substantially with data size, and that balanced sampling across taxa is essential for achieving high macro-level performance.

## 2.2 Genomic Foundation Models

Genomic foundation models (GFMs) are large-scale neural networks pre-trained on vast corpora of DNA sequences using self-supervised learning objectives. Inspired by the success of language models such as BERT [17] and GPT [18], GFMs treat DNA sequences as a “language” of nucleotides and learn contextual representations that capture biological structure and function.

### 2.2.1 Nucleotide Transformer

The Nucleotide Transformer (NT) [9] family of models, developed by InstaDeep, are among the largest and most widely evaluated GFMs. This thesis uses the `nucleotide-transformer-v2-500m-multi-species` variant, which features:

- **Architecture:** An ESM-2-based [19] Transformer encoder with 29 layers, 1,024 hidden dimensions, and 16 attention heads.
- **Tokenization:** A 6-mer tokenizer that encodes DNA sequences into non-overlapping 6-nucleotide tokens. This produces a vocabulary of  $4^6 = 4,096$  possible tokens plus special tokens.

- **Pre-training:** Masked language modeling (MLM) on 850 genomes from 174 species, spanning  $\sim 3.2$  billion nucleotides.

- **Total parameters:** 498M parameters.



The model produces contextual embeddings for each token in the input sequence, yielding a representation tensor of shape  $[\text{batch}, \text{seq\_len}, 1024]$  that captures both local sequence composition and long-range dependencies.

### 2.2.2 Other Genomic Foundation Models

**DNABERT** [10] was one of the first BERT-style models pre-trained on human genome sequences using overlapping  $k$ -mer tokenization ( $k = 3, 4, 5, 6$ ). While effective for tasks such as promoter prediction, its relatively small scale (110M parameters) and human-genome-only pre-training limit its applicability to diverse metagenomic sequences.

**DNABERT-2** [11] replaces  $k$ -mer tokenization with Byte Pair Encoding (BPE), eliminating the need to choose a fixed  $k$  and enabling more flexible sequence representation. It also adopts a multi-species pre-training strategy.

**HyenaDNA** [20] adopts single-nucleotide tokenization combined with sub-quadratic Hyena convolutions, enabling context windows up to 1M tokens at a fraction of the compute cost of attention-based models. Its single-character vocabulary eliminates the  $k$  design choice entirely, representing the opposite end of the tokenization design space from  $k$ -mer models.

**Caduceus** [21] is the first RC-equivariant bidirectional DNA language model. By constructing BiMamba layers with hard-coded reverse-complement weight sharing, Ca-

duceus enforces strand symmetry at the architectural level rather than relying on augmentation or post-hoc inference tricks. This is directly relevant to the RC consistency analysis in Section 2.5.1; unlike Caduceus, the NT-v2 backbone does not natively enforce RC equivariance, motivating our use of RC training augmentation and RC TTA.

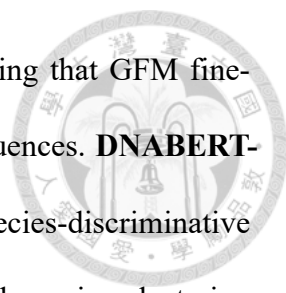
**Evo** [12] is a 7B-parameter model based on the StripedHyena architecture [22], capable of modeling DNA sequences up to 131K nucleotides. While powerful for long-range genomic modeling, its large size makes it challenging to deploy for metagenomic classification tasks involving hundreds of millions of short reads.

Across these models, a consistent theme emerges: tokenization strategy is a critical design axis. Lindsey, Fischman, et al. [23] systematically evaluate character, BPE,  $k$ -mer, Unigram, and WordPiece tokenizers across 44 genomic classification tasks and find accuracy differences of up to 5 percentage points depending on tokenizer choice, corroborating our own ablation results in Chapter 4.

### 2.2.3 GFMs for Metagenomic Classification

Despite their success on single-genome tasks, GFMs have seen limited application to metagenomic read-level classification. A naïve approach extracts mean-pooled embeddings from a frozen pre-trained model and trains a downstream classifier (e.g., MLP or SVM) on these fixed representations. While simple, this approach discards the token-level structure of the representation.

Several recent works have begun to close this gap, though none directly addresses genus-level taxonomic classification of short metagenomic reads at scale. **ViraLM** [24] fine-tunes DNABERT-2 for binary virus/non-virus classification of metagenomic contigs,



reporting a 22% F1 improvement on short fragments —demonstrating that GFM fine-tuning is beneficial even on compositionally diverse metagenomic sequences. **DNABERT-S** [25] adapts DNABERT-2 with contrastive learning to produce species-discriminative embeddings, achieving improvements on metagenomic binning and species clustering tasks; however, it operates at the embedding level rather than performing direct read classification. Both works fine-tune pre-existing GFMs and treat metagenomics as a secondary application; neither performs systematic tokenization ablation nor evaluates at the genus-level multi-class classification scale (120 genera, 50M reads) addressed in this thesis.

This thesis addresses the gap by proposing a token-level classification approach that preserves the full embedding sequence, fine-tunes the backbone with LoRA, and provides the first systematic comparison of pre-trained GFM fine-tuning against from-scratch Meta-Transformer training across tokenization strategies on the same large-scale dataset.

## 2.3 Parameter-Efficient Fine-Tuning

Fine-tuning the full parameter set of a 498M-parameter model requires substantial compute and memory, and risks catastrophic forgetting of the pre-trained representations. Parameter-efficient fine-tuning (PEFT) methods address this by adapting only a small subset of parameters while keeping the majority frozen.

### 2.3.1 Low-Rank Adaptation (LoRA)

LoRA [13] is based on the hypothesis that weight updates during fine-tuning lie in a low-rank subspace. For a pre-trained weight matrix  $W_0 \in \mathbb{R}^{d \times k}$ , LoRA parameterizes the



update as:

$$W = W_0 + \Delta W = W_0 + BA \quad (2.1)$$

where  $B \in \mathbb{R}^{d \times r}$  and  $A \in \mathbb{R}^{r \times k}$  with  $\text{rank } r \ll \min(d, k)$ . During training, only  $A$  and  $B$  are updated while  $W_0$  remains frozen. The forward pass becomes:

$$h = W_0 x + \frac{\alpha}{r} B A x \quad (2.2)$$

where  $\alpha$  is a scaling hyperparameter that controls the magnitude of the adaptation.

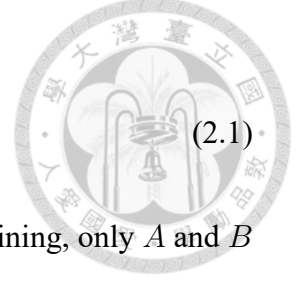
Key advantages of LoRA include:

- **Parameter efficiency:** With  $r = 16$  applied to query, key, and value projections across 29 Transformer layers, only 5.54M parameters (1.11% of total) are trainable.
- **No additional inference latency:** The adapted weights  $W_0 + BA$  can be merged, adding no overhead at inference time.
- **Modular:** Different LoRA adapters can be trained for different tasks and swapped at deployment time.

### 2.3.2 Other PEFT Methods

**Adapter layers** [26] insert small trainable bottleneck modules between Transformer layers. While effective, they introduce additional inference latency due to the extra forward-pass computation.

**Prefix tuning** [27] prepends learnable “virtual tokens” to the input sequence, influencing the attention computation without modifying model weights. It is memory-efficient



but can struggle with tasks requiring fine-grained output control.

We choose LoRA for this thesis due to its combination of strong performance, parameter efficiency, and the absence of additional inference latency.



## 2.4 Attention Mechanisms for Sequence Aggregation

Given a sequence of token embeddings  $\{h_1, h_2, \dots, h_T\} \in \mathbb{R}^{T \times d}$ , a classification task requires aggregating these into a fixed-dimensional representation. The choice of aggregation mechanism directly impacts what information is preserved.

### 2.4.1 Mean Pooling

The simplest approach averages all token embeddings:

$$z = \frac{1}{T} \sum_{t=1}^T h_t \quad (2.3)$$

Mean pooling is computationally trivial but weights all tokens equally, potentially diluting the contribution of discriminative tokens with non-informative ones.

### 2.4.2 Attention Pooling

Attention pooling uses learnable query vectors to compute a weighted combination of token embeddings. Given  $M$  learnable query vectors  $\{q_1, \dots, q_M\} \in \mathbb{R}^{M \times d_q}$  and linear

projections  $W_K, W_V \in \mathbb{R}^{d \times d_q}$ :

$$K = HW_K, \quad V = HW_V \quad (2.4)$$

$$\alpha_{m,t} = \frac{\exp(q_m \cdot k_t / \sqrt{d_q})}{\sum_{t'} \exp(q_m \cdot k_{t'} / \sqrt{d_q})} \quad (2.5)$$

$$z_m = \sum_{t=1}^T \alpha_{m,t} v_t \quad (2.6)$$

$$z = \text{Concat}(z_1, \dots, z_M) \in \mathbb{R}^{M \cdot d_q} \quad (2.7)$$



This mechanism enables the model to learn which positions in the DNA sequence are most informative for classification, automatically focusing on discriminative  $k$ -mers while down-weighting uninformative regions.

### 2.4.3 Multi-Head Attention Pooling

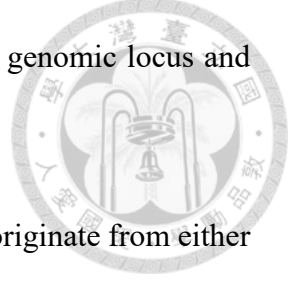
Extending attention pooling with multiple heads allows the model to attend to different aspects of the sequence simultaneously. Each head learns a different query, capturing diverse patterns (e.g., one head may focus on conserved marker genes while another attends to variable regions). The concatenated outputs from all heads provide a rich, multi-faceted representation for downstream classification.

## 2.5 Data Augmentation for DNA Sequences

### 2.5.1 Reverse-Complement Symmetry in DNA

DNA is a double-stranded molecule: the two antiparallel strands are linked by Watson–Crick base pairing ( $A \leftrightarrow T, C \leftrightarrow G$ ). Given a sequence  $s = s_1 s_2 \dots s_L$  on the forward ( $5' \rightarrow 3'$ ) strand, its *reverse complement*  $\bar{s} = \bar{s}_L \dots \bar{s}_1$  on the complementary strand en-

codes identical biological information: both correspond to the same genomic locus and therefore the same organism.



In shotgun metagenomic sequencing, a read is equally likely to originate from either strand. Consequently, a biologically correct classifier must satisfy the *strand symmetry* property:

$$f(s) = f(\bar{s}) \quad \forall s \quad (2.8)$$

However, standard neural network architectures do not inherently enforce this constraint. Zhou et al. [28] systematically demonstrate that deep learning models for genomics frequently violate strand symmetry, producing inconsistent predictions for a sequence and its reverse complement. Ma [29] further show that even DNA language models fine-tuned with RC data augmentation still exhibit substantial prediction inconsistency.

## 2.5.2 Strategies for Exploiting RC Symmetry

Three complementary strategies have been proposed to address the strand symmetry problem:

1. **Random RC training augmentation:** During training, each read is replaced with its reverse complement with probability  $p = 0.5$ , exposing the model to both strand orientations. While this encourages—but does not guarantee—strand-invariant representations, it effectively doubles the training data diversity at zero storage cost.
2. **RC test-time augmentation (RC TTA):** At inference, both the forward read  $s$  and its reverse complement  $\bar{s}$  are processed independently, and their output logits are

averaged before taking the argmax prediction:

$$\hat{y} = \arg \max_k [f_\theta(s) + f_\theta(\bar{s})] \quad (2.9)$$



This “post-hoc conjoined” approach [28] enforces strand symmetry at inference time without modifying the trained model. It can be understood as a two-view ensemble where both views observe the same biological entity from different strand orientations. By the standard ensemble argument, averaging two correlated but noisy estimates of the same ground-truth label reduces prediction variance and cancels orientation-dependent errors. Zhou et al. [28] find that this approach consistently performs well across diverse genomic tasks.

3. **RC consistency training:** A regularization loss (e.g., KL divergence) explicitly penalizes divergence between  $f_\theta(s)$  and  $f_\theta(\bar{s})$  during training, producing a single model that is intrinsically more strand-invariant without requiring double inference at test time [29]. An alternative is to design *RC-equivariant architectures* that hard-code strand symmetry into the network structure [30], though this requires custom layers incompatible with pre-trained foundation model backbones.

In this thesis, we adopt strategies (1) and (2): RC augmentation during training and RC TTA during evaluation. We also experiment with RC consistency training (Section 4.4) but find that its interaction with other regularization techniques (e.g., logit adjustment) can be detrimental, and RC TTA alone provides a reliable +1–1.5 percentage point improvement across all experimental settings (Section 4.7).



## 2.6 Handling Class Imbalance

Metagenomic samples exhibit severe class imbalance: a few dominant taxa may constitute the majority of reads while rare taxa contribute only a handful. This imbalance can bias classifiers toward frequent classes, resulting in high micro accuracy but poor macro F1 due to frequent misclassification of rare taxa.

### 2.6.1 Loss-Level Approaches

**Class-weighted cross-entropy** scales the loss for each class inversely proportional to its frequency:

$$\mathcal{L} = - \sum_{k=1}^K w_k y_k \log \hat{y}_k, \quad w_k \propto 1/n_k \quad (2.10)$$

However, this approach can be unstable with mixed-precision training and large class counts [31].

**Logit adjustment** [32] modifies the output logits by a class-prior-dependent offset:

$$\tilde{f}_k(x) = f_k(x) + \tau \log \pi_k \quad (2.11)$$

where  $\pi_k$  is the prior probability of class  $k$  and  $\tau$  is a temperature parameter controlling the strength of adjustment. This approach is theoretically motivated as producing Fisher-consistent estimators under class imbalance.



### 2.6.2 Sampling-Level Approaches

**Balanced sampling** constructs each training batch by sampling classes uniformly and then sampling instances within each class. This ensures that rare classes are seen as frequently as common ones during training.

**Class-aware sampling** (e.g., `WeightedRandomSampler`) assigns sampling weights inversely proportional to class frequency, achieving a similar effect to balanced batches within standard data loading pipelines.

## 2.7 Summary

This chapter has reviewed the key concepts underpinning our proposed approach: metagenomic classification methods from alignment-based to deep learning, genomic foundation models that provide rich pre-trained representations, parameter-efficient fine-tuning via LoRA, attention-based sequence aggregation, and strategies for data augmentation and class imbalance. In the next chapter, we describe how these components are integrated into a cohesive classification framework.

# Chapter 3

## Methodology



This chapter describes the proposed token-level genomic foundation model classification framework. We present the overall system architecture, detail each component, and describe the training strategy, data pipeline, and evaluation methodology.

### 3.1 System Overview

Figure 3.1 illustrates the end-to-end pipeline. Given a raw DNA read of 150 bp, the system performs the following steps:

1. **Tokenization:** The read is segmented into non-overlapping 6-mer tokens using the Nucleotide Transformer v2 tokenizer.
2. **Token-level encoding:** The token sequence is processed by the pre-trained Nucleotide Transformer v2 backbone (with LoRA adapters), producing contextual embeddings for each token.
3. **Attention pooling:** A multi-head attention pooling head aggregates the token embeddings into a fixed-dimensional classification vector.
4. **Classification:** A feed-forward network maps the pooled representation to class logits.



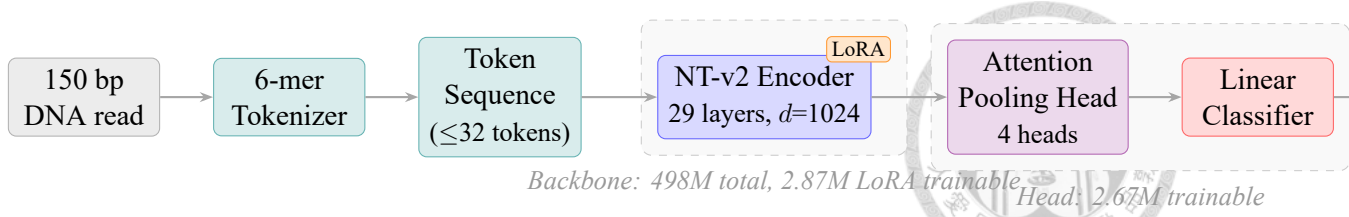
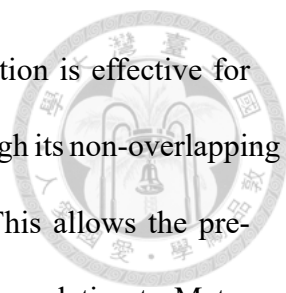


Figure 3.1: Overview of the proposed token-level GFM classification pipeline. A 150 bp DNA read is tokenized into non-overlapping 6-mers, encoded by the LoRA-adapted NT-v2 backbone (29 layers,  $d=1024$ ), aggregated via multi-head attention pooling, and classified into one of  $K$  taxonomic classes (120 genera or 1,535 species). Orange badge indicates LoRA adapter modules; dashed boxes delineate the backbone and classification head parameter groups.

## 3.2 Backbone: Nucleotide Transformer v2

We employ the nucleotide-transformer-v2-500m-multi-species model as the backbone encoder. This model is based on the ESM-2 architecture [19], a protein language model architecture adapted for nucleotide sequences. The selection of NT-v2 over other genomic foundation models is motivated by four considerations:

1. **Empirical performance on metagenomic reads:** A direct comparison of frozen-backbone baselines (Section 4.2) shows that NT-v2 achieves 51.70% genus-level accuracy, outperforming the 7B-parameter Evo-2 model (46.95%) despite having  $14\times$  fewer parameters. This result provides direct empirical evidence that NT-v2’s representations are better suited to metagenomic read classification.
2. **Multi-species pre-training:** NT-v2’s multi-species variant is pre-trained on 850 genomes spanning 174 species [9], making its pre-training distribution the closest among available GFMs to the multi-organism nature of metagenomic data. By contrast, DNABERT [10] is pre-trained exclusively on the human genome, which is taxonomically far from gut microbial sequences.
3.  **$k$ -mer tokenization as a common ground with MetaTransformer:** MetaTrans-



former [8] demonstrates that  $k$ -mer-based sequence discretization is effective for metagenomic classification. NT-v2 shares this philosophy through its non-overlapping 6-mer tokenizer, establishing a direct methodological link. This allows the pre-trained 6-mer embeddings to serve as a principled initialization relative to Meta-Transformer’s learned-from-scratch 12-mer embeddings. DNABERT-2 [11], while multi-species, replaces  $k$ -mer tokenization with BPE, breaking this alignment.

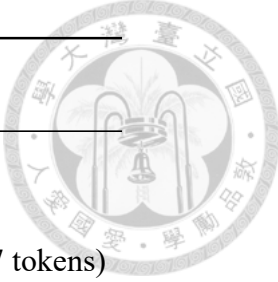
4. **Practical scalability:** At 498M parameters, NT-v2 can be fine-tuned with LoRA on a single H100 GPU with well under 10 GB of VRAM, making it feasible to run hundreds of training epochs over tens of millions of reads. Evo’s 7B parameters [12], while powerful for long-range genomic modeling, impose substantially higher memory requirements that are difficult to justify for 150 bp short-read classification.

### 3.2.1 Tokenization

**Stride and overlap.** A  $k$ -mer tokenizer slides a window of length  $k$  along a DNA sequence, converting each window into a vocabulary token. The *stride*  $s$  controls the step size between consecutive windows:

- **Non-overlapping** ( $s = k$ ): each nucleotide belongs to exactly one token. A 150 bp read yields  $\lfloor 150/k \rfloor$  tokens.
- **Overlapping** ( $s = 1$ ): every nucleotide position starts a new token. A 150 bp read yields  $150 - k + 1$  tokens.

For example, a 12-nucleotide sequence AGTCGATTCGCA tokenized with  $k=6$  gives:



| Stride                  | Tokens                                 |
|-------------------------|----------------------------------------|
| $s=6$ (non-overlapping) | AGTCGA TTCGCA (2 tokens)               |
| $s=1$ (overlapping)     | AGTCGA, GTCGAT, TCGATC, ... (7 tokens) |

Table 3.1 summarises the resulting token counts for a 150 bp read under the configurations used in this thesis.

Table 3.1: Token counts for a 150 bp read under different  $k$ -mer tokenization settings.

| $k$ | Stride $s$ | Overlap         | Tokens/read |
|-----|------------|-----------------|-------------|
| 6   | 6          | Non-overlapping | 25          |
| 6   | 1          | Overlapping     | 145         |
| 12  | 12         | Non-overlapping | 12          |
| 12  | 1          | Overlapping     | 139         |

**NT-v2 tokenization.** The NT-v2 tokenizer uses non-overlapping 6-mers ( $k=6$ ,  $s=6$ ), the setting used during its pre-training. For a 150 bp read this produces 25 tokens (plus [CLS] and [EOS] special tokens). The maximum sequence length is set to 128 tokens, accommodating reads up to  $\sim 768$  bp with padding. Section 4.11 presents a controlled ablation that quantifies the effect of different tokenization settings on classification accuracy.

### 3.2.2 Encoder Architecture

The backbone consists of 29 Transformer encoder layers, each with:

- Multi-head self-attention with 16 heads and hidden dimension 1,024.
- A feed-forward network (FFN) with intermediate dimension 4,096.
- Pre-layer normalization (LayerNorm before attention and FFN).

- Rotary position embeddings (RoPE) [33] for encoding positional information.

The backbone outputs a tensor  $H \in \mathbb{R}^{B \times T \times 1024}$ , where  $B$  is the batch size and  $T$  is the sequence length. We use the *full token sequence* (excluding special tokens) as input to the classification head, rather than extracting only the [CLS] token or applying mean pooling.

### 3.2.3 LoRA Adaptation

We apply LoRA [13] to the query ( $W_Q$ ), key ( $W_K$ ), and value ( $W_V$ ) projection matrices in each of the 29 self-attention layers. The LoRA configuration is:

Table 3.2: LoRA hyperparameters.

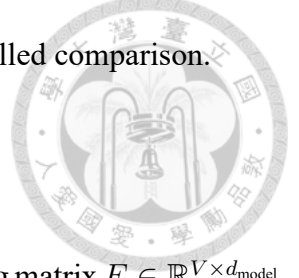
| Parameter                  | Value             |
|----------------------------|-------------------|
| Rank $r$                   | 16                |
| Scaling factor $\alpha$    | 32                |
| Dropout                    | 0.05              |
| Target modules             | $W_Q, W_K, W_V$   |
| Trainable LoRA params      | 2,867,200         |
| Classification head params | 2,672,248         |
| Total trainable params     | 5,539,448 (1.11%) |
| Total model params         | 498,623,561       |

Gradient checkpointing is enabled for the backbone to reduce peak GPU memory usage at the cost of recomputing activations during the backward pass.

### 3.2.4 Ablation: Shallow Transformer Backbone

To isolate the contribution of the pre-trained 29-layer NT-v2 backbone, we design an ablation variant that replaces it with a shallow Transformer encoder trained entirely from scratch. This ablation mirrors the architecture of MetaTransformer [8] while keeping the

tokenization fixed to 6-mers (the NT-v2 tokenizer), enabling a controlled comparison.



The shallow Transformer backbone consists of:

- **Learned token embeddings:** A randomly initialized embedding matrix  $E \in \mathbb{R}^{V \times d_{\text{model}}}$ , where  $V = 4,107$  is the NT-v2 6-mer vocabulary size and  $d_{\text{model}} = 128$ .
- **Learned positional embeddings:** A position embedding matrix  $P \in \mathbb{R}^{L_{\text{max}} \times d_{\text{model}}}$ , added to the token embeddings.
- **Single-layer Transformer encoder:** One pre-norm Transformer encoder layer with 2 attention heads, feed-forward dimension  $d_{\text{ff}} = 512$ , GELU activation, and dropout  $p = 0.1$ .

Table 3.3: Comparison of backbone configurations.

| Aspect           | NT-v2 + LoRA          | Shallow (ablation)    |
|------------------|-----------------------|-----------------------|
| Tokenizer        | 6-mer (shared)        | 6-mer (shared)        |
| Encoder layers   | 29                    | 1                     |
| Hidden dimension | 1,024                 | 128                   |
| Attention heads  | 16                    | 2                     |
| Feed-forward dim | 4,096                 | 512                   |
| Pre-training     | MLM on 850 genomes    | None (from scratch)   |
| Backbone params  | 498M (5.5M trainable) | ~200K (all trainable) |

The same attention pooling classification head is used for both variants, with the input dimension adjusted to match the backbone output ( $d = 1,024$  for NT-v2,  $d = 128$  for the shallow variant). This design ensures that any performance difference is attributable solely to the backbone encoder, not the classification head or tokenization.

The shallow variant trains all parameters from scratch, analogous to MetaTransformer’s training paradigm but constrained to the NT-v2 tokenizer. This answers a key question: *does the pre-trained 29-layer backbone provide an advantage over a simple shallow encoder when the tokenization is held constant?*

### 3.3 Classification Head: Attention Pooling



The classification head aggregates the variable-length token sequence into a fixed-dimensional representation and maps it to class logits. We use a multi-head attention pooling mechanism inspired by Perceiver [34] and Set Transformer [35].

#### 3.3.1 Architecture

The attention pooling head consists of:

1. **Learnable query tokens:**  $M = 4$  learnable query vectors  $Q \in \mathbb{R}^{M \times d}$ , where  $d = 1024$ .
2. **Cross-attention:** Standard scaled dot-product attention where the queries attend to the backbone's token embeddings:

$$K = HW_K, \quad V = HW_V \quad (3.1)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (3.2)$$

where  $H \in \mathbb{R}^{T \times d}$  is the backbone output and  $W_K, W_V \in \mathbb{R}^{d \times d_k}$  are learnable projections.

3. **Aggregation:** The  $M$  attended vectors are concatenated, yielding a representation of dimension  $M \times d_k$ .
4. **Feed-forward classifier:**

$$\text{LayerNorm} \rightarrow \text{Linear}(M \cdot d_k, 512) \rightarrow \text{GELU} \rightarrow \text{Dropout}(p) \rightarrow \text{Linear}(512, K) \quad (3.3)$$

where  $K$  is the number of classes.



The attention pooling mechanism enables the model to selectively focus on the most taxonomically informative tokens in the sequence, analogous to how a biologist might focus on specific conserved marker regions within a read to determine its taxonomic origin.

### 3.3.2 Alternative Heads

We also implemented two alternative classification heads for comparison:

**Mean Pool Head:** Averages all token embeddings along the sequence dimension, followed by the same feed-forward classifier. This serves as the baseline aggregation strategy.

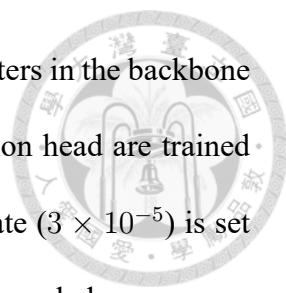
**Transformer Pool Head:** Applies a single Transformer encoder layer on top of the backbone output before mean pooling, adding another level of self-attention. This explores whether an additional attention layer can compensate for simple mean pooling.

## 3.4 Training Strategy

### 3.4.1 Two-Phase Training

We employ a two-phase training strategy to stabilize learning:

**Phase 1 — Head-only warmup (5 epochs):** The backbone is completely frozen, and only the classification head parameters are trained. This phase initializes the head weights to produce meaningful gradients before the backbone is unfrozen, preventing early gradient noise from corrupting pre-trained representations.



**Phase 2 — Joint fine-tuning** (up to 25–30 epochs): LoRA adapters in the backbone are unfrozen, and both the backbone (via LoRA) and the classification head are trained jointly. Differential learning rates are used: the backbone learning rate ( $3 \times 10^{-5}$ ) is set lower than the head learning rate ( $5 \times 10^{-4}$ ) to preserve pre-trained knowledge.

### 3.4.2 Optimization

The training procedure uses the following optimization configuration:

Table 3.4: Training optimization hyperparameters.

| Parameter                   | Value                     |
|-----------------------------|---------------------------|
| Optimizer                   | AdamW                     |
| Backbone learning rate      | $3 \times 10^{-5}$        |
| Head learning rate          | $5 \times 10^{-4}$        |
| Weight decay                | 0.02                      |
| LR scheduler                | Cosine with linear warmup |
| Warmup ratio                | 10% of total steps        |
| Gradient accumulation steps | 1–2                       |
| Mixed precision             | bfloat16                  |

**Mixed precision training:** We use bfloat16 automatic mixed precision (AMP) on NVIDIA H100 GPUs. The bfloat16 format provides the dynamic range of float32 with reduced precision, avoiding the gradient underflow issues that arise with float16 and class-weighted losses.

**Early stopping:** Training is halted when validation accuracy does not improve for 7 consecutive epochs, and the model with the highest validation accuracy is saved as the final checkpoint.





### 3.4.3 Transfer Learning Across Experiments

To facilitate progressive scaling from smaller to larger datasets, we implement a transfer learning mechanism (`--init_from`) distinct from the training resumption mechanism (`--resume`).

**Resume** (`--resume`): Loads the full training state (model weights, optimizer state, epoch counter, learning rate scheduler) from a checkpoint, enabling continuation of the *same* experiment after a job interruption (e.g., SLURM time limit).

**Transfer learning** (`--init_from`): Loads only the model weights from a checkpoint for a *new* experiment, resetting the epoch counter and optimizer state. Supports partial loading, automatically skipping parameters with shape mismatches (e.g., when the classification head dimension changes between genus and species models). Phase 1 warmup is skipped when using transfer learning, as the backbone is already adapted.

## 3.5 Data Pipeline

### 3.5.1 Training Data Generation

Training data consists of simulated metagenomic reads generated from reference genomes using the ART simulator [36], following the same protocol as MetaTransformer [8]. Reference genomes are drawn from the Human Gastrointestinal Reference genome database (HGR-UMGS) [37], covering 120 genera and 1,535 species of human gut-associated microorganisms.

Each simulated read is 150 bp in length (matching typical Illumina HiSeq output) and is labeled with its source organism's genus and species.



### 3.5.2 Balanced Subsampling

The full training dataset contains 258M reads with a highly skewed distribution across species. To mitigate class imbalance and enable controlled data scaling experiments, we implement a balanced subsampling strategy:

1. **Scan:** The full FASTA file is scanned to extract all read headers and their species labels.
2. **Allocate:** A target number of reads per species is computed as  $n_{\text{target}} = N_{\text{total}} / K_{\text{species}}$ , where  $N_{\text{total}}$  is the desired subsample size and  $K_{\text{species}}$  is the number of species.
3. **Cap and redistribute:** For species with fewer reads than  $n_{\text{target}}$ , all available reads are included. The deficit is redistributed proportionally among species with surplus reads.
4. **Sample:** Reads are randomly selected within each species to meet the per-species allocation.

This procedure produces near-perfectly balanced subsamples. For a 50M subsample from 1,535 species, each species contributes  $32,573 \pm 445$  reads.

### 3.5.3 Reverse-Complement Augmentation

During training, each read is replaced with its reverse complement with probability  $p = 0.5$ . This augmentation is applied on-the-fly during data loading, doubling the effective diversity of the training set without requiring additional storage.



### 3.5.4 Data Splitting

The dataset is split into training (90%) and validation (10%) sets using stratified sampling to maintain class proportions. For species-level experiments, stratification ensures that every species is represented in both splits.

## 3.6 Evaluation Methodology

### 3.6.1 Metrics

We evaluate classification performance using the following metrics:

**Micro accuracy** (overall accuracy): The fraction of correctly classified reads. This metric is dominated by abundant classes and reflects real-world performance on typical samples.

**Macro F1**: The unweighted average of per-class F1 scores. This metric weights all classes equally, providing a measure of performance across the full taxonomic diversity including rare taxa.

**Balanced accuracy**: The average of per-class recall, equivalent to macro recall.

**Top- $k$  accuracy**: The fraction of reads for which the correct class appears in the top  $k$  predictions. We report top-3, top-5, and top-10 accuracy.

**F1=0 class count**: The number of classes with zero F1 score (i.e., classes for which the model makes no correct predictions). This directly measures the “dead class” problem.

**Train-validation gap**: The difference between training accuracy and validation accuracy, serving as a proxy for overfitting.

### 3.6.2 Reverse-Complement Test-Time Augmentation

At inference, we employ RC TTA: each read is classified twice (forward and reverse complement), and the logits are averaged before applying softmax:

$$\hat{y} = \arg \max_k [f_{\theta}(r) + f_{\theta}(\bar{r})] \quad (3.4)$$

where  $f_{\theta}(r)$  and  $f_{\theta}(\bar{r})$  are the logit vectors for the forward and reverse-complement reads, respectively. This ensembling exploits the biological strand symmetry of DNA and consistently improves accuracy.

### 3.6.3 Species-Level Evaluation Modes

For species-level classification (1,535 classes), we introduce two additional evaluation modes to decouple the contributions of genus-level and intra-genus discrimination:

**Oracle-genus evaluation:** Given the *true* genus label, the species logits are masked to only allow species belonging to that genus. This answers: “Given perfect genus identification, how well can the model discriminate species within each genus?”

**Top- $k$  genus routing:** The genus classifier produces the top- $k$  genus predictions. The candidate species set is the union of all species belonging to these  $k$  genera. Species logits are then masked to this candidate set. This simulates a realistic two-stage hierarchical classification pipeline.





## 3.7 Computational Considerations

### 3.7.1 Hardware

All experiments are conducted on the TWCC (Taiwan Computing Cloud) SLURM cluster equipped with NVIDIA H100 80 GB HBM3 GPUs. The SLURM partition enforces a 48-hour maximum wall time per job, necessitating checkpoint-resume support for long training runs.

### 3.7.2 Memory Optimization

Several techniques are employed to manage GPU memory:

- **Gradient checkpointing:** Recomputes intermediate activations during the backward pass rather than storing them, trading  $\sim 30\%$  additional compute for  $\sim 60\%$  memory reduction.
- **Mixed precision (bfloat16):** Reduces memory requirements for activations and gradients by approximately 50% compared to float32.
- **LoRA:** Keeps the majority of backbone parameters frozen, reducing the optimizer state memory from  $\sim 4\text{GB}$  (full fine-tuning with AdamW) to  $\sim 44\text{MB}$ .

### 3.7.3 Resource Monitoring

To assess computational resource adequacy and identify bottlenecks, we implement a dedicated resource monitoring module that tracks:

- GPU VRAM usage (PyTorch allocated, nvidia-smi total, and peak values).

- GPU compute utilization (sampled every 30 seconds via `nvidia-smi`).
- System RAM usage (process RSS and peak).
- Per-epoch wall-clock time and throughput (reads per second).



These metrics are logged to `training_history.csv` and a comprehensive `resource_report.json` at the end of training.

### 3.8 Summary

This chapter has described the complete methodology for token-level GFM classification of metagenomic reads. The framework combines a pre-trained Nucleotide Transformer v2 backbone with LoRA adaptation, an attention-pooling classification head, two-phase training with transfer learning support, balanced subsampling, and RC augmentation. The evaluation methodology encompasses multiple metrics at both genus and species levels, with specialized evaluation modes for dissecting hierarchical classification performance.



# Chapter 4

## Experiments and Results



This chapter presents the experimental setup and comprehensive results. We begin with frozen-embedding baselines, proceed through architecture and training ablations on a small dataset, and culminate in data scaling experiments that reveal data volume as the dominant factor in classification performance.

### 4.1 Experimental Setup

#### 4.1.1 Dataset

Training data is generated by simulating Illumina short reads (150 bp, HiSeq 2500 error profile) from the HGR-UMGS reference genome database [37] using the ART read simulator [36], following the same protocol as MetaTransformer [8]. The dataset covers:

- **120 genera** and **1,535 species** of human gut microorganisms.
- **258M total reads** across all species.
- Highly skewed distribution: the most abundant genus (*Collinsella*) accounts for 22.8% of reads, while the rarest genera contribute  $<0.06\%$ .

We conduct experiments at three data scales by subsampling from the full dataset:

- **500K** (imbalanced): The initial small-scale dataset preserving the natural class distribution. Train: 450K, validation: 50K.
- **5M** (balanced): Species-balanced subsample. Each of 1,535 species contributes



~3,257 reads. Train: 4.5M, validation: 500K.

- **50M (balanced)**: Species-balanced subsample. Each species contributes ~32,573 reads. Train: 45M, validation: 5M.



### 4.1.2 Evaluation Protocol

All models are evaluated on a held-out validation set using the metrics described in Section 3.6. Unless otherwise noted, results are reported with RC TTA (Eq. 3.4).

### 4.1.3 Hardware and Training Infrastructure

All experiments are run on NVIDIA H100 80 GB HBM3 GPUs on the TWCC SLURM cluster. The 48-hour SLURM time limit requires checkpoint-resume support for training runs exceeding this duration.

## 4.2 Phase 1: Frozen Embedding Baselines

Before developing the token-level approach, we establish baseline performance using precomputed, mean-pooled embeddings from frozen GFMs.

### 4.2.1 Setup

Embeddings are extracted once from each of several pre-trained GFMs. A two-layer MLP classifier is trained on the resulting fixed-dimensional vectors. We evaluate models from two families:

- **Evo-2 (7B params)**: A large-scale model based on the StripedHyena architecture.

We extract embeddings from layer 2.

- **Nucleotide Transformer v2 (500M params):** ESM-2-based model. We extract embeddings from the last hidden layer.



## 4.2.2 Results

Table 4.1: Frozen embedding baselines for genus classification (120 classes, 500K dataset).

| Method                                      | Genus Accuracy | Notes                     |
|---------------------------------------------|----------------|---------------------------|
| Uniform random ( $1/K$ )                    | 0.83%          | Lower bound               |
| Frequency baseline ( $\sum p_k^2$ )         | 8.11%          | Predicting proportionally |
| Majority class (always <i>Collinsella</i> ) | 22.84%         | Naïve baseline            |
| Evo-2 L2 mean-pool + MLP                    | 45.52%         |                           |
| Evo-2 L2 + Focal Loss                       | 46.95%         |                           |
| NT-v2 mean-pool + MLP                       | 51.70%         | Best frozen baseline      |
| NT-v2 mean-pool + MLP (RC-avg)              | 51.70%         | With RC averaging         |

The NT-v2 backbone consistently outperforms Evo-2 despite having  $14\times$  fewer parameters, suggesting that NT-v2’s multi-species pre-training provides more suitable representations for metagenomic classification. NT-v2 achieves 51.70% genus accuracy, which serves as our primary baseline.

## 4.3 Phase 2: Token-Level Classification with LoRA

We now present the token-level approach, progressing from initial architecture validation through systematic ablation studies.

### 4.3.1 Architecture Validation (v1–v3, 500K Dataset)

**Key findings from v1–v3:**

Table 4.2: Hyperparameter configurations for initial architecture experiments (500K dataset). Bold values indicate the setting that changed relative to v1.

| Hyperparameter  | v1                 | v2                     | v3 (best)          |
|-----------------|--------------------|------------------------|--------------------|
| Backbone LR     | $5 \times 10^{-5}$ | $2 \times 10^{-5}$     | $3 \times 10^{-5}$ |
| Head LR         | $5 \times 10^{-4}$ | $3 \times 10^{-4}$     | $5 \times 10^{-4}$ |
| RC augmentation | No                 | <b>Yes</b>             | <b>Yes</b>         |
| Class weights   | No                 | <b>Yes (capped@10)</b> | No                 |
| Label smoothing | 0.0                | <b>0.1</b>             | 0.0                |
| Weight decay    | 0.01               | 0.05                   | 0.02               |
| Head dropout    | 0.1                | 0.2                    | 0.15               |
| LoRA dropout    | 0.05               | 0.1                    | 0.05               |

Table 4.3: Results for architecture experiments (500K dataset, genus classification).

| Version | Val Accuracy  | F1 (weighted) | Best Epoch | Key Change              |
|---------|---------------|---------------|------------|-------------------------|
| v1      | 51.70%        | 0.4865        | 10/15      | Baseline                |
| v2      | 27.87%        | 0.3210        | 18/26      | Class weights (harmful) |
| v3      | <b>53.92%</b> | <b>0.5128</b> | 28/35      | RC augmentation         |

1. The token-level approach with LoRA (v1) matches the frozen mean-pool baseline (51.70%), validating the architecture but showing that token-level representations alone do not guarantee improvement when the backbone is partially frozen via LoRA.
2. Class-weighted cross-entropy with label smoothing (v2) catastrophically degrades performance to 27.87%, a  $-23.8$  pp drop. This is caused by the interaction between class weights and label smoothing on a highly imbalanced 120-class problem.
3. RC augmentation (v3) provides a solid  $+2.2$  pp improvement (51.70%  $\rightarrow$  53.92%), confirming the value of exploiting DNA strand symmetry.

### 4.3.2 Training Dynamics and Overfitting

The training dynamics reveal severe overfitting: while training accuracy continues to increase to 62.7%, validation accuracy saturates at  $\sim 54\%$ . The growing train-validation

Table 4.4: Training dynamics for v3 (500K dataset). The train-validation gap grows steadily, reaching 7% at the best epoch and 9% by epoch 35.

| Epoch | Train Acc | Val Acc | Train-Val Gap | Val Loss |
|-------|-----------|---------|---------------|----------|
| 1     | 45.25%    | 46.29%  | −1.0%         | 2.044    |
| 5     | 51.34%    | 51.27%  | +0.1%         | 1.809    |
| 10    | 54.25%    | 52.39%  | +1.9%         | 1.755    |
| 15    | 56.25%    | 53.26%  | +3.0%         | 1.735    |
| 20    | 58.08%    | 53.88%  | +4.2%         | 1.721    |
| 28*   | 60.88%    | 53.92%  | +7.0%         | 1.748    |
| 35    | 62.69%    | 53.71%  | +9.0%         | 1.762    |

Best validation accuracy epoch.

gap (0%  $\rightarrow$  9%) indicates that the model has sufficient capacity to memorize the training data but fails to generalize, suggesting that 500K reads is insufficient to train a 498M-parameter model, even with only 1.1% of parameters unfrozen.

Figure 4.1 visualises the training trajectories across all three data scales, illustrating this transition from severe overfitting (v3, 500K) to well-generalised training (v9, 50M).

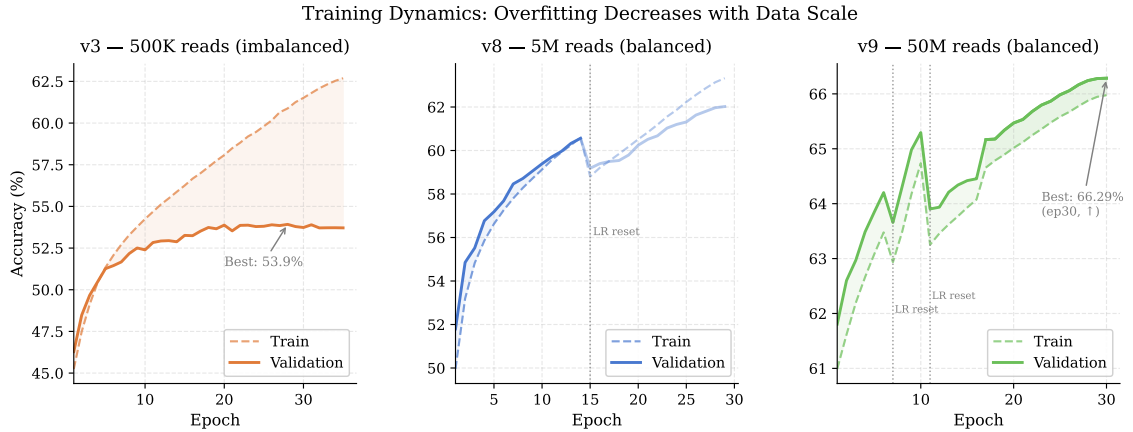


Figure 4.1: Training dynamics for v3 (500K, left), v8 (5M, centre), and v9 (50M, right). Dashed lines show training accuracy; solid lines show validation accuracy. The shaded region highlights the train-validation gap. Dotted vertical lines mark learning-rate resets caused by SLURM timeout and job resumption; v9 experienced two such resets (epochs 7 and 11) before the resume logic was corrected. Overfitting decreases systematically as data volume increases.

## 4.4 Advanced Training Techniques (v4–v7, 500K Dataset)

We explore several techniques to improve performance on the 500K dataset. Table 4.5 summarizes all experiments.

Table 4.5: Advanced training technique experiments (500K dataset, genus classification). RC TTA results are shown where available.

| Ver. | Description        | Fwd Acc | RC TTA | F1-macro | F1=0 | Gap   |
|------|--------------------|---------|--------|----------|------|-------|
| v3   | RC aug, maxlen=128 | 53.92%  | 55.36% | 25.82%   | 11   | 6.96% |
| v4   | maxlen=32          | 53.75%  | 55.29% | 25.70%   | 9    | 6.25% |
| v5   | LA $\tau=1.0$      | 35.17%  | —      | —        | —    | —     |
| v5b  | LA $\tau=0.3$      | 52.29%  | 53.58% | 26.53%   | 5    | —     |
| v6   | RC consist.+LA     | 35.87%  | 38.19% | 22.88%   | —    | —     |
| v7   | RC consist. only   | 54.10%  | 55.18% | 25.10%   | 11   | —     |

**Logit Adjustment** (v5, v5b): Logit adjustment (Eq. 2.11) with  $\tau = 1.0$  is too aggressive, dropping accuracy to 35.17%. A milder  $\tau = 0.3$  preserves micro accuracy (52.29%) while reducing F1=0 classes from 9 to 5, showing a favorable trade-off between overall and per-class performance.

**Reduced sequence length** (v4): Reducing `max_length` from 128 to 32 tokens slightly reduces accuracy ( $-0.07$  pp) but also reduces the overfitting gap (6.96%  $\rightarrow$  6.25%) and decreases training time.

**RC Consistency Training** (v6, v7): Adding a KL divergence loss between forward and reverse-complement predictions (v7) achieves 55.18% RC TTA, comparable to v3. However, combining RC consistency with logit adjustment (v6) degrades performance significantly, indicating that these regularization techniques interfere with each other.

**Conclusion:** All training techniques on the 500K dataset yield at most  $\pm 0.5$  pp improvement in RC TTA accuracy (range: 53.58%–55.36%). This plateau strongly suggests that *data volume*, not model architecture or training tricks, is the primary bottleneck.



## 4.5 Data Scaling Experiments

Motivated by the accuracy plateau on 500K data and the comparison with MetaTrans-former [8] (which trained on a large-scale balanced dataset trained on the full HGR-UMGS dataset of 2,505 species with coverage factor 4), we systematically scale training data.

### 4.5.1 Experiment v8: 5M Balanced Reads

#### 4.5.1.1 Setup

A 5M species-balanced subsample is drawn from the full 258M dataset. Each of 1,535 species contributes  $\sim 3,257$  reads, yielding near-perfect class balance. The v3 architecture and training configuration are retained (batch size 32, gradient accumulation 2, effective batch size 64), and training proceeds for 30 epochs.

Due to the 48-hour SLURM time limit, training required two jobs: the initial run completed 14 of 20 epochs before timeout, and a resumed run completed 16 additional epochs (epochs 15–30) using the `--resume` mechanism. A third job performed final evaluation.

#### 4.5.1.2 Results

The results reveal clear log-linear scaling behaviour:

1. **Accuracy:** Each  $10\times$  data increase yields a substantial but diminishing gain in RC TTA accuracy:  $+7.76$  pp (500K $\rightarrow$ 5M) and  $+4.02$  pp (5M $\rightarrow$ 50M). The sub-linear trend is consistent with logarithmic scaling laws.

Table 4.6: Genus classification results across data scales: v4 (500K), v8 (5M balanced), and v9 (50M balanced).

| Metric                  | v4 (500K) | v8 (5M) | $\Delta_{v4 \rightarrow v8}$ | v9 (50M) <sup>†</sup> | $\Delta_{v8 \rightarrow v9}$ |
|-------------------------|-----------|---------|------------------------------|-----------------------|------------------------------|
| Micro Accuracy (fwd)    | 53.75%    | 62.02%  | +8.27 pp                     | 66.29%                | +4.27 pp                     |
| Micro Accuracy (RC TTA) | 55.29%    | 63.05%  | +7.76 pp                     | <b>67.07%</b>         | +4.02 pp                     |
| Balanced Accuracy       | 25.23%    | 33.35%  | +8.12 pp                     | 39.91%                | +6.56 pp                     |
| F1 (weighted)           | 52.51%    | 61.06%  | +8.55 pp                     | 65.61%                | +4.55 pp                     |
| F1 (macro)              | 25.70%    | 37.97%  | +12.27 pp                    | <b>44.82%</b>         | +6.85 pp                     |
| Top-3 Accuracy          | —         | 81.70%  | —                            | 84.40%                | +2.70 pp                     |
| Top-5 Accuracy          | 82.91%    | 88.08%  | +5.17 pp                     | 90.03%                | +1.95 pp                     |
| Top-10 Accuracy         | 91.29%    | 94.23%  | +2.94 pp                     | 95.25%                | +1.02 pp                     |
| F1=0 classes            | 9         | 2       | −7                           | <b>0</b>              | −2                           |
| Train-Val Gap           | 6.25%     | 1.30%   | −4.95 pp                     | <1%                   | —                            |

<sup>†</sup> v9 final metrics (epoch 30, converged).

2. **Macro F1:** Data scaling disproportionately benefits rare classes: +12.27 pp at the first step and +6.85 pp at the second. At 50M reads, all 120 genera have non-zero F1 for the first time.
3. **F1=0 classes:** The number of unclassifiable genera drops from 9 to 2 (v8) and finally to 0 (v9), meaning every genus in the dataset is correctly predicted for at least some reads.
4. **Overfitting:** The train-validation gap shrinks dramatically from 6.25% (v4) to 1.30% (v8) to <1% (v9), confirming that the model transitions from data starvation to well-generalised training as data volume increases.

#### 4.5.1.3 Training Dynamics

The temporary accuracy drop at epoch 15 is caused by the learning rate scheduler resetting to its warmup phase after resumption. The model recovers within 3–4 epochs and eventually surpasses the pre-interruption performance.

Table 4.7: Training dynamics for v8 (5M balanced, 30 epochs). The gap between training and validation accuracy remains small throughout, in sharp contrast to the 500K experiments.

| Epoch | Train Acc | Val Acc | Train-Val Gap | Notes              |
|-------|-----------|---------|---------------|--------------------|
| 1     | 49.96%    | 51.80%  | -1.8%         |                    |
| 5     | 55.97%    | 57.04%  | -1.1%         |                    |
| 10    | 58.93%    | 59.42%  | -0.5%         |                    |
| 14    | 60.60%    | 60.56%  | +0.04%        | 48h limit          |
| 15    | 58.80%    | 59.18%  | -0.4%         | Resumed (LR reset) |
| 20    | 61.82%    | 60.59%  | +1.2%         |                    |
| 25    | 62.89%    | 61.77%  | +1.1%         |                    |
| 29*   | 63.32%    | 62.02%  | +1.3%         | Best epoch         |

## 4.5.2 Experiment v9: 50M Balanced Reads

Building on v8, we scale training data by another  $10\times$  to 50M balanced reads. The batch size is increased from 32 to 256 (with gradient accumulation reduced from 2 to 1, yielding an effective batch size of 256) to improve GPU utilization. The model is initialized from v8’s best checkpoint using `--init_from` (transfer learning mode).

Table 4.8: Configuration comparison between v8 and v9.

| Parameter             | v8 (5M)      | v9 (50M)       |
|-----------------------|--------------|----------------|
| Training reads        | 4.5M         | 45M            |
| Reads per species     | $\sim 3,257$ | $\sim 32,573$  |
| Batch size            | 32           | 256            |
| Gradient accumulation | 2            | 1              |
| Effective batch size  | 64           | 256            |
| Epochs                | 30           | 30 (converged) |
| Weight initialization | Random       | v8 best.pt     |
| Est. time per epoch   | $\sim 3h$    | $\sim 5-6h$    |

### 4.5.2.1 Results

Scaling from 5M to 50M balanced reads yields a further +4.02 pp improvement in RC TTA accuracy (63.05%  $\rightarrow$  67.07%). Critically, the number of F1=0 classes drops to **zero**, indicating that all 120 genera are now correctly classified in at least some reads.



Table 4.9: Comparison between v8 (5M) and v9 (50M) for genus classification (120 classes).

| Metric                  | v8 (5M) | v9 (50M)       | $\Delta$ |
|-------------------------|---------|----------------|----------|
| Micro Accuracy (fwd)    | 62.02%  | <b>66.29%</b>  | +4.27 pp |
| Micro Accuracy (RC TTA) | 63.05%  | <b>67.07%</b>  | +4.02 pp |
| Balanced Accuracy       | 33.35%  | 39.91%         | +6.56 pp |
| F1 (weighted)           | 61.06%  | 65.61%         | +4.55 pp |
| F1 (macro)              | 37.97%  | <b>44.82%</b>  | +6.85 pp |
| Top-3 Accuracy          | 81.70%  | 84.40%         | +2.70 pp |
| Top-5 Accuracy          | 88.08%  | 90.03%         | +1.95 pp |
| Top-10 Accuracy         | 94.23%  | 95.25%         | +1.02 pp |
| F1=0 classes            | 2       | <b>0</b>       | −2       |
| Best epoch              | 29      | 30 (converged) | —        |

The marginal return diminishes relative to the previous  $10\times$  scaling step:  $500K \rightarrow 5M$  yielded +7.76 pp while  $5M \rightarrow 50M$  yields +4.02 pp for the same multiplicative data increase, consistent with a logarithmic scaling relationship. This sub-linear trend is expected under log-linear scaling laws and suggests that continued data scaling alone will not close the remaining gap with MetaTransformer.

**Training progress and LR restart incidents.** v9 training required multiple SLURM jobs due to the 48-hour time limit. Before a bug fix was applied to the checkpoint-resume logic, two job restarts inadvertently reset the cosine learning-rate scheduler to its initial warmup phase, causing temporary accuracy dips: −0.54 pp at epoch 7 and −1.39 pp at epoch 11. In both cases the model recovered within 3–5 epochs and surpassed the pre-dip performance level. The underlying issue—`EarlyStopping.best` being silently set to `None` from an incomplete checkpoint key—was corrected in the resume logic; from epoch 17 onward the scheduler state is correctly restored and the learning rate continues its cosine decay without interruption.

Training completed at epoch 30 (April 2026), reaching a forward validation accuracy

of **66.29%** and RC TTA accuracy of **67.07%** (Figure 4.2). The learning curve plateaued between epochs 26 and 30 as the cosine schedule reached its minimum, indicating convergence.

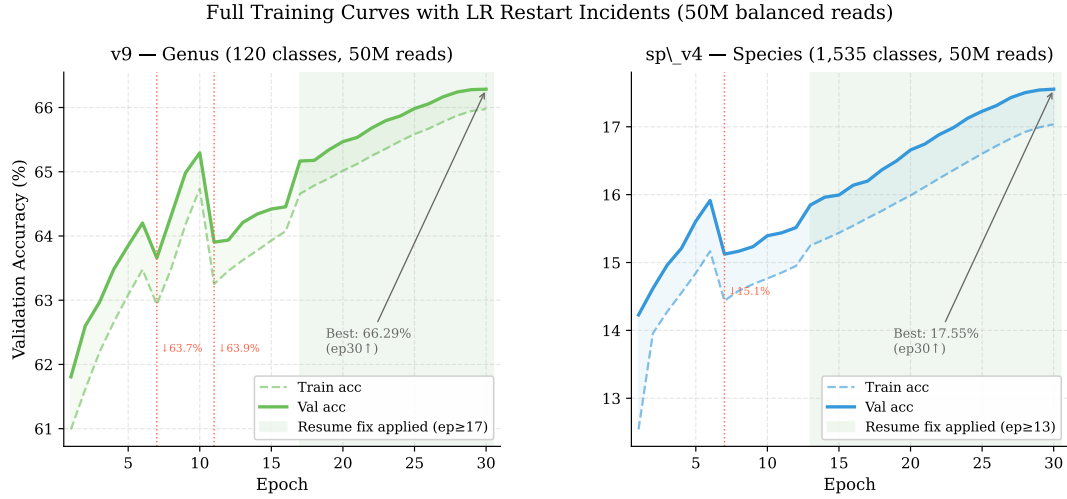


Figure 4.2: Full validation accuracy curves for v9 (genus, left) and sp\_v4 (species, right) over all completed epochs. Red dotted vertical lines mark epochs where a SLURM job restart incorrectly reset the cosine LR scheduler, causing a temporary accuracy dip. The green shaded region indicates epochs trained after the resume bug fix was applied, during which the LR schedule correctly continues its cosine decay.

### 4.5.3 Data Scaling Analysis

Table 4.10: Data scaling analysis: genus classification performance across data volumes.

| Data Volume                     | RC TTA Acc    | F1-macro      | F1=0     | Gap   |
|---------------------------------|---------------|---------------|----------|-------|
| 500K (imbalanced)               | 55.29%        | 25.70%        | 9        | 6.25% |
| 5M (balanced)                   | 63.05%        | 37.97%        | 2        | 1.30% |
| 50M (balanced)                  | <b>67.07%</b> | <b>44.82%</b> | <b>0</b> | ~0.7% |
| MetaTransformer (HGR-UMGS full) | ~98%          | —             | —        | —     |

The data scaling analysis reveals a clear relationship between training data volume and classification performance. The transition from 500K to 5M reads ( $10\times$  increase) yields +7.76 pp in accuracy and +12.27 pp in macro F1. Meanwhile, all training technique ablations on the 500K dataset (v3–v7) produced at most  $\pm 0.5$  pp variation. This strongly supports the conclusion that **data volume, not model architecture or training**

techniques, is the primary determinant of classification accuracy in this setting.

The comparison with MetaTransformer further reinforces this finding: despite using a much larger backbone (498M vs.  $\sim 5$ M parameters), our model at 5M training reads achieves only 63% accuracy, while MetaTransformer, trained on the full HGR-UMGS dataset with coverage factor 4, achieves 98.3% genus recall (98.9% precision) on its validation set. The gap is attributable primarily to the substantial difference in training data volume.

Figure 4.3 plots genus RC TTA accuracy against training data volume on a logarithmic scale, with a log-linear fit to our three data points. The fitted curve projects that closing the gap with MetaTransformer would require training data well beyond our 50M-read scale, consistent with diminishing returns under logarithmic scaling.

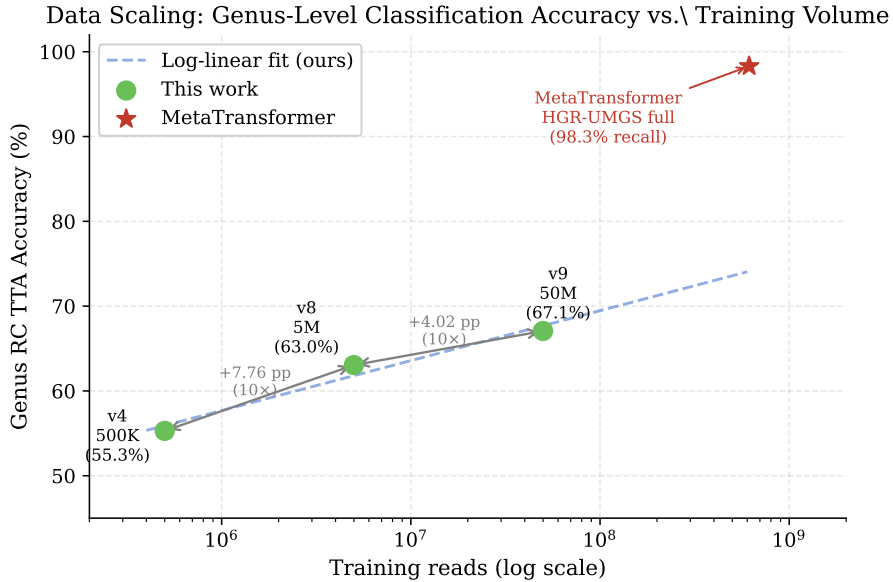


Figure 4.3: Genus RC TTA accuracy versus training data volume (log scale). Blue circles mark our results at three data scales; the red star marks MetaTransformer’s reported genus recall. The dashed line is a log-linear fit to our three data points. Per-10 $\times$  accuracy gains diminish from +7.76 pp (500K $\rightarrow$ 5M) to +4.02 pp (5M $\rightarrow$ 50M). The MetaTransformer star marks reported genus recall (98.3%); its x-position is based on training throughput (300K steps  $\times$  batch 2,048) and is not directly comparable to our dataset-size axis.



## 4.6 Species-Level Classification

We evaluate species-level classification (1,535 classes) using the 500K dataset to establish baselines and understand the difficulty of fine-grained taxonomic discrimination.

### 4.6.1 Direct Classification

Table 4.11: Species classification results (500K dataset, 1,535 classes).

| Version | Description      | RC TTA Acc | Oracle Acc | F1-macro | F1=0 |
|---------|------------------|------------|------------|----------|------|
| sp_v1   | Baseline         | 8.32%      | 20.88%     | —        | 613  |
| sp_v2   | LA $\tau=0.3$    | 8.37%      | 20.85%     | 20.0%    | 433  |
| sp_v3   | Weighted sampler | 8.01%      | 20.48%     | 19.9%    | 392  |

Direct species classification achieves only  $\sim 8\%$  accuracy, which, while far above the random baseline (0.065% for 1,535 classes), indicates that 500K reads is wholly insufficient for fine-grained species discrimination.

### 4.6.2 Oracle-Genus Evaluation

Oracle-genus evaluation reveals that even with perfect genus knowledge, species-level accuracy reaches only  $\sim 21\%$ . This ceiling indicates that the model struggles with *intra-genus* discrimination—distinguishing species within the same genus—which is expected given the high sequence similarity among congeners at the 150 bp read length.

### 4.6.3 Effect of Class Imbalance Mitigation

Logit adjustment ( $\tau = 0.3$ , sp\_v2) and weighted sampling (sp\_v3) both substantially reduce the number of F1=0 classes (from 613 to 433 and 392, respectively) with minimal

impact on overall accuracy ( $< 0.4$  pp). This demonstrates that class imbalance mitigation is valuable for improving the breadth of species covered, even when it does not improve aggregate accuracy.



#### 4.6.4 Top- $k$ Genus Routing

Using top-10 genus routing, species accuracy reaches 8.06%, slightly below the direct classification result (8.32%). This modest result reflects the genus model’s limited accuracy at 500K scale (55%): when the correct genus is not in the top-10, the species classifier is working from an incorrect candidate set.

### 4.7 RC Test-Time Augmentation Analysis

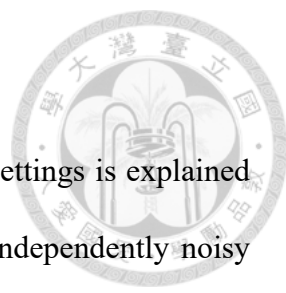
Table 4.12: Effect of RC TTA across experiments. RC TTA consistently improves accuracy at the cost of  $2\times$  inference time.

| Experiment                 | Fwd Only | RC TTA | $\Delta$ |
|----------------------------|----------|--------|----------|
| v3 (500K genus)            | 53.92%   | 55.36% | +1.44 pp |
| v4 (500K genus)            | 53.75%   | 55.29% | +1.54 pp |
| v5b (500K, LA $\tau=0.3$ ) | 52.29%   | 53.58% | +1.29 pp |
| v7 (500K, RC consist.)     | 54.10%   | 55.18% | +1.08 pp |
| v8 (5M genus)              | 62.02%   | 63.05% | +1.03 pp |
| v9 (50M genus, NT-v2)      | 66.29%   | 67.07% | +0.78 pp |
| v11 (50M genus, shallow)   | 53.80%   | 53.88% | +0.08 pp |

RC TTA provides a consistent +1.0 to +1.5 pp accuracy improvement across all experiments, regardless of the training configuration or data scale. Several observations merit discussion:

1. **Biological basis:** As discussed in Section 2.5.1, a metagenomic read and its reverse complement encode identical biological information. RC TTA exploits this strand symmetry by treating the two orientations as a two-view ensemble of the same un-

derlying genomic entity [28].

- 
2. **Variance reduction:** The consistent improvement across all settings is explained by standard ensemble theory: averaging two correlated but independently noisy estimates of the same ground-truth label reduces prediction variance, even when both estimates come from the same model.
  3. **Diminishing returns with RC consistency training:** v7 (RC consistency, +1.08 pp) shows less benefit than vanilla models, because RC consistency training already encourages  $f(s) \approx f(\bar{s})$  during training, leaving less residual strand asymmetry for RC TTA to correct.
  4. **Scale-dependent diminishing returns:** The RC TTA benefit decreases slightly with data scale: +1.44 pp at 500K, +1.03 pp at 5M, and +0.78 pp at 50M (v9). Larger training sets expose the model to both strands of each locus more often, gradually reducing systematic strand-directional errors—though a non-trivial asymmetry persists even at 50M reads.
  5. **Near-zero benefit for randomly initialized models:** v11 (shallow Transformer, random initialization) shows only +0.08 pp from RC TTA. Without pre-training on a large genomic corpus, the model does not develop systematic strand-direction preferences, leaving little strand asymmetry for RC TTA to correct. This further supports the view that RC TTA primarily corrects pre-training-induced strand biases rather than stochastic inference noise.

The  $2\times$  inference cost is the primary trade-off. For offline evaluation this is acceptable; for latency-sensitive deployment, RC-equivariant architectures [30] or RC consistency regularization [29] may be preferred alternatives.

Figure 4.4 summarises RC TTA gains across all experiments, highlighting the near-zero benefit for the randomly initialized v11 model.

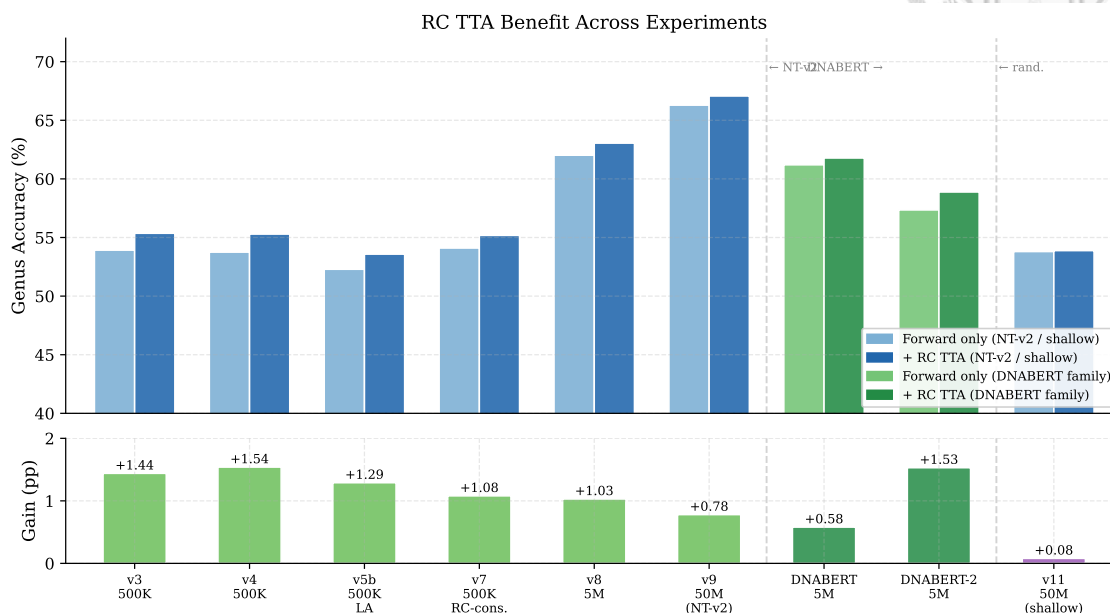


Figure 4.4: RC TTA benefit across experiments. Top: forward-only vs. RC TTA accuracy (blue: NT-v2 backbone; green: DNABERT family; right of second dashed line: random init). Bottom: absolute gain in percentage points. NT-v2-based models gain +0.77–+1.54 pp consistently. DNABERT gains only +0.58 pp, likely because its overlapping 6-mer tokenization is inherently more RC-symmetric than NT-v2’s non-overlapping 6-mer. DNABERT-2 (BPE tokenization) gains +1.53 pp, similar to early NT-v2 experiments. The randomly initialized v11 gains only +0.08 pp, confirming that the benefit primarily corrects pre-training-induced strand biases.

## 4.8 Computational Resource Analysis

### 4.8.1 Observed Resource Usage

### 4.8.2 Bottleneck Analysis

The hardware is substantially over-provisioned for this workload: the H100 80 GB GPU runs at <50% VRAM utilization even with batch size 256, and system RAM usage is well within limits. The primary operational bottleneck is the 48-hour SLURM wall time limit, which necessitates multiple job submissions with checkpoint resumption for large-scale experiments.

Table 4.13: Computational resource usage across experiments.

| Metric               | v3 (500K) | v8 (5M)    | v9 (50M)       |
|----------------------|-----------|------------|----------------|
| GPU                  | V100 32GB | H100 80GB  | H100 80GB      |
| Batch size           | 32        | 32         | 256            |
| VRAM peak (training) | 10.4 GB   | ~4.7 GB    | ~5 GB          |
| VRAM utilization     | 32.5%     | 5.5%       | <7%            |
| Train throughput     | 1,128 r/s | ~1,800 r/s | ~1,825 r/s     |
| Inference throughput | —         | ~5,100 r/s | ~5,170 r/s     |
| Epoch time           | ~6.5 min  | ~6.8h      | ~6.9h          |
| Total training       | 11.5h     | ~90h       | ~207h (30 ep.) |
| SLURM jobs needed    | 1         | 3          | 4+             |

Table 4.14: Resource bottleneck assessment on TWCC H100 nodes.

| Resource                | Status                    | Assessment                              |
|-------------------------|---------------------------|-----------------------------------------|
| GPU VRAM (80 GB)        | <50% utilized             | No upgrade needed                       |
| GPU compute (H100)      | Adequate                  | No upgrade needed                       |
| System RAM (128 GB)     | <50% utilized             | No upgrade needed                       |
| Storage                 | <10 GB per experiment     | No upgrade needed                       |
| <b>SLURM time (48h)</b> | <b>Primary bottleneck</b> | <b>Requires resume or longer limits</b> |

### 4.8.3 Scaling Projections

For the full 258M dataset training:

- **Memory:** 258M reads require ~52 GB RAM when loaded in-memory, which fits within the 128 GB system memory. Alternatively, a streaming data loader can reduce memory requirements to  $O(\text{batch size})$ .
- **Training time:** At the v8 throughput of ~417 reads/sec (batch size 32), one epoch over 232M training reads would take ~155 hours. With batch size 256 and improved GPU utilization, this could potentially be reduced to ~20–30 hours per epoch, requiring 2–3 SLURM jobs for 10 epochs.



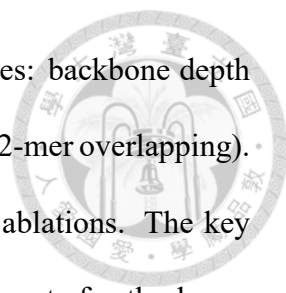
Table 4.15: Comparison between our approach and MetaTransformer [8].

| Aspect           | Ours (v9)              | MetaTransformer                       |
|------------------|------------------------|---------------------------------------|
| Backbone         | NT-v2 (498M)           | Custom Transformer ( $\sim 5$ M)      |
| Tokenization     | 6-mer (pre-trained)    | 12-mer (learned)                      |
| Pre-training     | Yes (850 genomes, MLM) | No (trained from scratch)             |
| Trainable params | 5.54M (LoRA)           | $\sim 5$ M (all)                      |
| Training data    | 50M balanced           | HGR-UMGS full (2,505 sp., coverage 4) |
| Genus accuracy   | 67.07% (RC TTA)        | 98.3% recall (val)                    |
| Data source      | HGR-UMGS + ART         | HGR-UMGS + ART (same)                 |

## 4.9 Comparison with MetaTransformer

Despite using a foundation model  $100\times$  larger than MetaTransformer, our model (at 50M training reads) substantially underperforms MetaTransformer (trained on the full HGR-UMGS dataset). The key differences are:

1. **Training data volume:** MetaTransformer uses the full HGR-UMGS dataset with coverage 4 vs. our 50M balanced reads. Given our data scaling results (each  $10\times$  data increase yields +3–8 pp with diminishing returns), this is likely the dominant factor.
2. **Training paradigm:** MetaTransformer trains all parameters from scratch ( $\sim 5$ M), while we fine-tune only 5.54M LoRA parameters of a 498M-parameter model. The foundation model’s pre-trained representations, while beneficial as a starting point, may not be optimally suited for metagenomic classification without sufficient fine-tuning data.
3. **Tokenization:** MetaTransformer uses 12-mer tokenization (capturing longer sequence patterns per token), while NT-v2 uses 6-mers. The coarser 12-mer tokenization may be advantageous for species discrimination, as it directly captures longer characteristic motifs.



This comparison is confounded by three simultaneous differences: backbone depth (29 vs. 1 layer), pre-training (yes vs. no), and tokenization (6-mer vs. 12-mer overlapping). Sections 4.10 and 4.11 disentangle these factors through controlled ablations. The key finding is that tokenization—specifically, overlapping 12-mer—accounts for the larger share of the performance gap, with pre-training providing a secondary but still substantial benefit under matched tokenization.

## 4.10 Backbone Ablation: Shallow Transformer

To isolate the contribution of the pre-trained 29-layer NT-v2 backbone from the effect of tokenization, we replace the backbone with a single-layer Transformer trained from scratch while keeping the 6-mer tokenizer and attention pooling head identical (see Section 3.2.4).

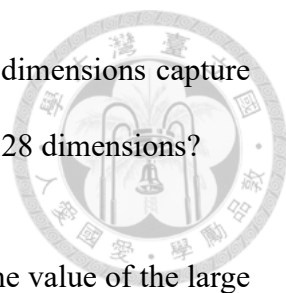
### 4.10.1 Experimental Design

Table 4.16: Controlled ablation: NT-v2 backbone (v9) vs. shallow Transformer (v11). All other components are held constant.

| Component           | v9 (NT-v2 + LoRA)        | v11 (Shallow)            |
|---------------------|--------------------------|--------------------------|
| Tokenizer           | 6-mer (NT-v2)            | 6-mer (NT-v2)            |
| Encoder             | 29-layer, $d=1024$       | 1-layer, $d=128$         |
| Pre-training        | MLM on 850 genomes       | None                     |
| Trainable params    | 5.54M (LoRA)             | ~200K (all)              |
| Classification head | Attention pool (4 heads) | Attention pool (4 heads) |
| Training data       | 50M balanced             | 50M balanced             |
| Batch size          | 256                      | 512                      |

This controlled comparison isolates two factors:

1. **Pre-trained representations:** Does pre-training on 850 genomes via masked language modeling provide an advantage for metagenomic classification?

- 
2. **Model capacity:** Does a 29-layer encoder with 1,024 hidden dimensions capture more discriminative features than a single-layer encoder with 128 dimensions?

If the shallow variant substantially underperforms, it validates the value of the large pre-trained backbone. If performance is comparable, it suggests that the pre-trained representations provide limited benefit for this task under the 6-mer tokenization constraint, and that a lightweight task-specific model may be sufficient.

#### 4.10.2 Expected Outcomes

Based on MetaTransformer’s success with a similarly shallow architecture (albeit with 12-mer tokenization and a much larger training set), we hypothesize:

- The shallow variant will underperform v9 at the same data scale (50M), because the 6-mer tokenizer produces shorter token sequences than 12-mers, requiring more encoder capacity to capture equivalent patterns.
- The performance gap will quantify the “value of pre-training” under fixed tokenization, informing whether future work should invest in larger backbones or better tokenizers.

#### 4.10.3 Results

The pre-trained 29-layer NT-v2 backbone outperforms the single-layer randomly initialized Transformer by **13.19 pp** in RC TTA accuracy (67.07% vs. 53.88%). This gap confirms that the pre-trained representations, despite being fine-tuned via only 5.54M LoRA parameters, provide substantial discriminative value that a shallow randomly initialized

Table 4.17: Backbone ablation results: pre-trained NT-v2 (v9) vs. shallow Transformer (v11), both trained on 50M balanced reads.

| Metric                  | v9 (NT-v2 + LoRA) | v11 (Shallow) | $\Delta$            |
|-------------------------|-------------------|---------------|---------------------|
| Micro Accuracy (fwd)    | 66.29%            | 53.80%        | −12.49 pp           |
| Micro Accuracy (RC TTA) | <b>67.07%</b>     | 53.88%        | −13.19 pp           |
| Balanced Accuracy       | 39.91%            | 22.28%        | −17.63 pp           |
| F1 (macro)              | 44.82%            | 25.78%        | −19.04 pp           |
| F1=0 classes            | 0                 | 3             | +3                  |
| RC TTA gain             | +0.78 pp          | +0.08 pp      | —                   |
| Best epoch              | 30 / 30           | 15 / 15       | —                   |
| Inference throughput    | ~5,170 r/s        | ~750,000 r/s  | $\approx 145\times$ |
| VRAM (training)         | ~5 GB             | ~0.07 GB      | —                   |

model cannot replicate even with 50M training reads.

Several additional observations are noteworthy:

1. **F1 gap is larger than accuracy gap:** The macro F1 gap (−19.04 pp) substantially exceeds the accuracy gap (−13.19 pp), indicating that v11 disproportionately fails on rare genera. This is consistent with v11’s lower capacity for learning subtle sequence features distinguishing rare taxa.
2. **RC TTA nearly ineffective for v11:** The shallow, randomly initialized model shows only +0.08 pp from RC TTA (vs. +0.77 pp for v9), supporting the interpretation that RC TTA primarily corrects systematic strand biases acquired during pre-training rather than stochastic inference noise.
3. **Efficiency trade-off:** v11 is  $\sim 145\times$  faster at inference (750K vs. 5,170 reads/sec) and uses  $\sim 70\times$  less VRAM during training. For latency-sensitive applications where 54% accuracy is acceptable, v11 represents a viable lightweight alternative.
4. **Pre-training vs. tokenization:** This experiment isolates pre-training while holding tokenization constant (6-mer), yielding a +13.19 pp pre-training advantage. Sec-

tion 4.11 further shows that switching to overlapping 12-mer tokenization within the MetaTransformer architecture yields +31.83 pp—nearly  $2.5\times$  larger than the pre-training effect. The remaining gap between v11 (54%) and MetaTransformer on the full dataset is therefore attributable primarily to tokenization design, with training data volume as a secondary factor.

Figure 4.5 presents a direct visual comparison of v9 and v11 across five evaluation metrics, quantifying the benefit of the pre-trained backbone across all dimensions.

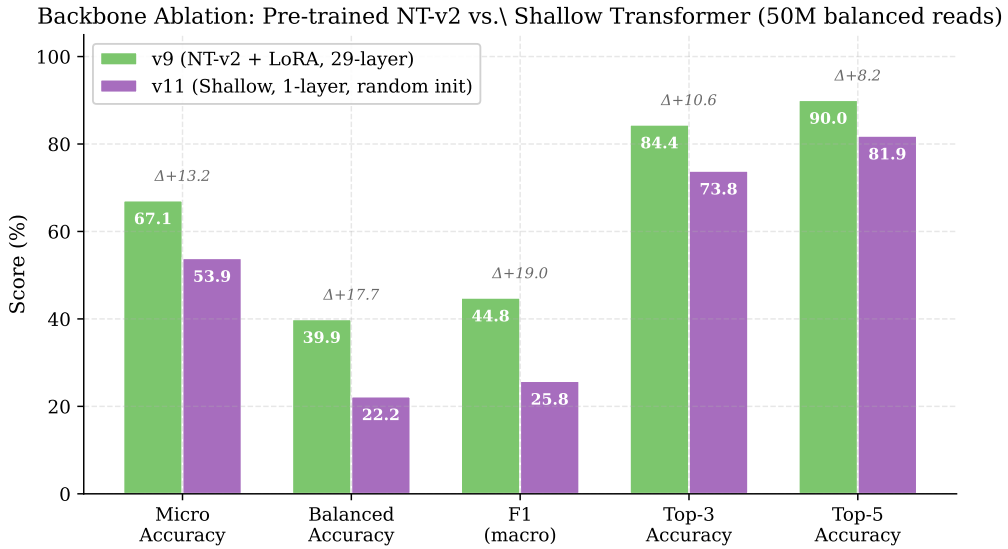


Figure 4.5: Backbone ablation: pre-trained NT-v2 + LoRA (v9, green) vs. single-layer randomly initialized Transformer (v11, purple), both trained on 50M balanced reads. Numbers above bars indicate the absolute gain of v9 over v11. The pre-trained backbone consistently outperforms the shallow variant across all metrics, with the largest gap in F1 (macro) (+19.0 pp) and Balanced Accuracy (+17.6 pp), reflecting v9’s superior discrimination of rare genera.

## 4.11 Tokenization Ablation: MetaTransformer at 50M Scale

The backbone ablation (Section 4.10) isolates the contribution of pre-training while holding tokenization fixed at 6-mer. To separately quantify the contribution of *tokenization*—the other major design difference between our approach and MetaTransformer—we train the MetaTransformer architecture [8] directly on the same 50M balanced dataset un-

der six tokenization configurations ( $k \in \{6, 12, 13\}$ ,  $\text{stride} \in \{1, k\}$ ), using the original MetaTransformer codebase on the Taiwania-2 cluster (V100 32 GB GPU). All configurations are evaluated on the same stratified 90/10 val split (5M reads). The 13-mer model uses  $\text{embed\_dim} = 64$  to fit within V100 memory, consistent with the original MetaTransformer paper’s 13-mer configuration [8].

Table 4.18: Tokenization ablation (genus-level, 120 classes): NT-v2 + LoRA vs. MetaTransformer under six tokenization configurations. All MetaTransformer models are trained from scratch on 50M balanced reads. RC TTA is unavailable for MetaTransformer.

| Model             | Tokenization      | Tokens/read | Top-1 Acc           | Macro F1      |
|-------------------|-------------------|-------------|---------------------|---------------|
| MetaTransformer   | 13-mer, stride=1  | 138         | <b>87.42%</b>       | <b>0.7936</b> |
| MetaTransformer   | 12-mer, stride=1  | 139         | 78.83%              | 0.6531        |
| NT-v2 + LoRA (v9) | 6-mer, stride=6   | 25          | 67.07% <sup>†</sup> | —             |
| NT-v2 + LoRA (v9) | 6-mer, stride=6   | 25          | 66.29%              | 0.4384        |
| MetaTransformer   | 6-mer, stride=1   | 145         | 48.87%              | 0.1981        |
| MetaTransformer   | 6-mer, stride=6   | 25          | 47.00%              | 0.1682        |
| MetaTransformer   | 12-mer, stride=12 | 12          | 40.66%              | 0.0663        |
| MetaTransformer   | 13-mer, stride=13 | 11          | 27.30%              | 0.0182        |

<sup>†</sup> RC TTA result (+0.78 pp over forward-only 66.29%); MetaTransformer results are forward-only.

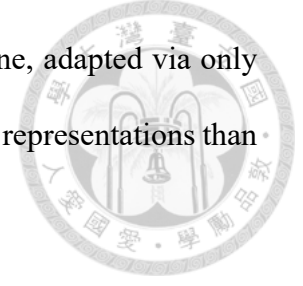
Table 4.19: Tokenization ablation (species-level, 1,535 classes): MetaTransformer under six tokenization configurations, trained on the same 50M balanced reads.

| Model           | Tokenization      | Tokens/read | Top-1 Acc     | Macro F1      |
|-----------------|-------------------|-------------|---------------|---------------|
| MetaTransformer | 13-mer, stride=1  | 138         | <b>49.62%</b> | <b>0.4950</b> |
| MetaTransformer | 12-mer, stride=1  | 139         | 16.79%        | 0.1439        |
| MetaTransformer | 6-mer, stride=1   | 145         | 9.13%         | 0.0719        |
| MetaTransformer | 6-mer, stride=6   | 25          | 6.44%         | 0.0470        |
| MetaTransformer | 12-mer, stride=12 | 12          | 0.64%         | 0.0023        |
| MetaTransformer | 13-mer, stride=13 | 11          | 0.51%         | 0.0026        |

Three independent effects can be read from Tables 4.18–4.19:

**Effect of pre-training (6-mer tokenization held constant).** NT-v2 + LoRA (66.29% forward) outperforms MetaTransformer with the same non-overlapping 6-mer tokenization (47.00%) by **+19.29 pp**. Holding architecture and tokenization fixed, this isolates the

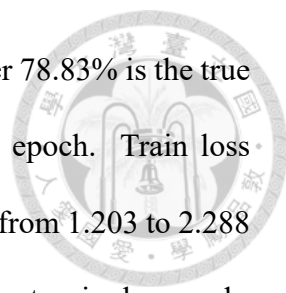
contribution of large-scale genomic pre-training: the NT-v2 backbone, adapted via only 5.54M LoRA parameters, provides substantially more discriminative representations than a 5M-parameter model trained from scratch on identical data.



**Effect of tokenization (MetaTransformer architecture held constant).** Tokenization design has a far larger effect than pre-training. Two sub-effects compound:

1. **Stride (overlapping vs. non-overlapping):** Across all  $k$  values, overlapping stride=1 dominates non-overlapping stride=  $k$ . The gap widens dramatically with  $k$ : +1.87 pp for  $k = 6$ , +38.17 pp for  $k = 12$ , and +60.12 pp for  $k = 13$ . Non-overlapping tokenization with large  $k$  reduces each 150 bp read to as few as 11 tokens, discarding most sequence context.
2.  **$k$ -mer length (overlapping only):** Among stride=1 configurations, longer  $k$  yields monotonically higher accuracy: 6-mer (48.87%) < 12-mer (78.83%) < 13-mer (87.42%) at genus level, and 9.13% < 16.79% < 49.62% at species level. Longer tokens are taxonomically more specific—hexamers are shared across distant taxa, while 12-mers and 13-mers increasingly identify individual lineages.

**Tokenization surpasses pre-training: MetaTransformer 13-mer vs. NT-v2.** MetaTransformer with overlapping 13-mer tokenization, trained *from scratch* on 45M reads, achieves **87.42%** genus Top-1 accuracy—**+20.35 pp** above NT-v2 + LoRA with RC TTA (67.07%), despite having  $\sim 100\times$  fewer parameters and no pre-training. At species level, MT 13-mer stride=1 reaches **49.62%** (1,535 classes) without any hierarchical routing. This result demonstrates that task-specific tokenization design is a more decisive factor than pre-trained prior knowledge for metagenomic read classification.



**Convergence and capacity ceiling (12-mer).** To determine whether 78.83% is the true ceiling for the 12-mer model, training was extended into a second epoch. Train loss dropped sharply from 0.764 to 0.119, while validation loss *worsened* from 1.203 to 2.288—immediate overfitting. The 50M-read dataset saturates the 5M-parameter single-encoder MetaTransformer within one epoch.

**Overlapping tokenization within pre-trained GFMs (5M scale).** The MetaTransformer ablation above is conducted entirely from scratch at 50M scale. To isolate the effect of overlapping tokenization *within the pre-trained GFM family*, we additionally fine-tune two other pre-trained DNA language models on the same 5M balanced dataset using identical training settings (LoRA  $r = 16$ , attention-pooling head):

- **DNABERT** [10]: BERT-base (86M params) pre-trained on the human reference genome with overlapping 6-mer tokenization (stride=1, 145 tokens per 150 bp read).
- **DNABERT-2** [11]: BERT-base (117M params) pre-trained on a multi-species genome collection using learned BPE tokenization ( $\sim 34$  tokens per read).

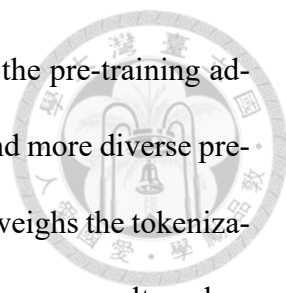
Table 4.20: Pre-trained GFM backbone comparison at 5M balanced reads (genus-level, 120 classes). All models use LoRA  $r = 16$  fine-tuning and the same attention-pooling head. RC TTA is applied to all models.

| Model             | Tokenization                  | Tokens/read | Params | RC TTA Acc    | Macro      |
|-------------------|-------------------------------|-------------|--------|---------------|------------|
| NT-v2 + LoRA (v8) | 6-mer, stride=6 (non-overlap) | 25          | 499M   | <b>63.05%</b> | <b>0.4</b> |
| DNABERT + LoRA    | 6-mer, stride=1 (overlap)     | 145         | 86M    | 61.78%        | 0.3        |
| DNABERT-2 + LoRA  | BPE ( $\sim 34$ tokens)       | $\sim 34$   | 117M   | 58.88%        | 0.3        |

Three observations follow from Table 4.20.

First, NT-v2 with *non-overlapping* 6-mer tokenization outperforms DNABERT with



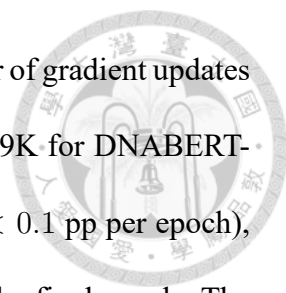


*overlapping* 6-mer tokenization by +1.27 pp. At 5M training reads, the pre-training advantage from NT-v2’s larger backbone (499M vs. 86M parameters) and more diverse pre-training corpus (850 microbial genomes vs. human genome only) outweighs the tokenization benefit of overlapping stride. This contrasts with the MetaTransformer results: when starting from scratch on the same data, overlapping tokenization provides a large benefit (+1.87 pp at  $k = 6$ , +60.12 pp at  $k = 13$ ); but when starting from a strong pre-trained representation, the non-overlapping NT-v2 backbone already captures sufficient local context through its positional encodings.

Second, DNABERT with overlapping 6-mer tokenization (pre-trained, 86M, 5M data, 61.78%) outperforms MetaTransformer with the same overlapping 6-mer tokenization (from scratch, 5M params, 50M data, 48.87%), despite ten times less training data. This confirms that pre-training on genomic sequences provides a substantial data-efficiency advantage—even a smaller pre-trained model is more data-efficient than a larger randomly initialised model.

Third, DNABERT-2’s BPE tokenization (58.88%) performs worse than either 6-mer model despite having a similar parameter count to DNABERT. BPE segments DNA by frequency-based byte pairs rather than fixed-length k-mers, which may fragment taxonomically informative motifs; the learned vocabulary may not align well with the k-mer statistics that distinguish metagenomic reads.

**Methodological note on training equivalence.** The three models use the same learning rate ( $5 \times 10^{-4}$  head/LoRA,  $3 \times 10^{-5}$  backbone), warmup ratio (5%), cosine decay schedule, and epoch count (30), but differ in batch size (NT-v2: 32, DNABERT: 128, DNABERT-2: 256) due to GPU memory constraints imposed by sequence length differences (25, 145,



and  $\sim 34$  tokens per read respectively). Consequently, the total number of gradient updates differs: approximately 453K for NT-v2, 117K for DNABERT, and 59K for DNABERT-2. All three learning curves were still improving at epoch 30 (gain  $< 0.1$  pp per epoch), with the cosine schedule decaying the learning rate to near zero by the final epoch. The comparison is therefore epoch-based—each model has seen the training data 30 times—rather than step-based or compute-matched. This is standard practice in fine-tuning comparisons, and the observed ranking (NT-v2  $>$  DNABERT  $>$  DNABERT-2) is unlikely to be reversed by additional training: DNABERT, despite receiving  $3.9\times$  fewer gradient updates than NT-v2, lags by only 1.27 pp, while DNABERT-2’s gap of 4.17 pp is consistent across all training stages and attributable primarily to its BPE tokenization.

**Implication for NT-v2.** NT-v2 is constrained to 6-mer tokenization by its pre-training design. The ablation results show that this constraint is the primary limitation of our approach: switching to overlapping 13-mer tokenization on the same architecture and data yields a +20.35 pp improvement without any pre-training. A genomic foundation model pre-trained with overlapping large- $k$  tokenization on diverse microbial genomes would combine the discriminative power of longer k-mers with the representational quality of large-scale pre-training. No such model currently exists publicly.

## 4.12 Species-Level Classification at Scale

Following the data scaling analysis for genus classification, we extend the 50M balanced training to species-level classification (1,535 classes).

Table 4.21: Species classification scaling: 500K (sp\_v1–v3) vs. 50M (sp\_v4).

| Parameter             | sp_v1–v3 (500K) | sp_v4 (50M)      |
|-----------------------|-----------------|------------------|
| Training reads        | 450K            | 45M              |
| Reads per species     | ~293            | ~29,316          |
| Classes               | 1,535           | 1,535            |
| Batch size            | 32              | 256              |
| Weight initialization | Random          | v9 genus best.pt |

### 4.12.1 Experimental Design

The species model is initialized from the v9 genus model’s backbone weights via transfer learning (`--init_from`), which automatically loads matching LoRA parameters while skipping the genus-specific classification head layers (120-class  $\rightarrow$  1,535-class dimension mismatch). This allows the species model to benefit from the genus-tuned backbone representations.

### 4.12.2 Final Results (sp\_v4, Epoch 30)

sp\_v4 training converged at epoch 30 (April 2026), reaching a forward validation accuracy of **17.55%** on the full 5M held-out set (1,535 classes). Table 4.22 shows the progression across epochs; Table 4.23 presents the complete metric profile at convergence.

Table 4.22: sp\_v4 training progress (species classification, 1,535 classes, 50M reads).

| Epoch | Val Acc (fwd) | Val F1 (macro) | Note                       |
|-------|---------------|----------------|----------------------------|
| 1     | 14.23%        | 12.43%         | Warm-up                    |
| 6     | 15.91%        | 14.05%         | End of 1 <sup>st</sup> job |
| 7     | 15.12%        | 13.36%         | LR reset (−0.79 pp)        |
| 12    | 15.51%        | 13.65%         | End of 2 <sup>nd</sup> job |
| 13    | 15.85%        | 13.97%         | Resume fix applied         |
| 18    | 16.37%        | 14.46%         |                            |
| 26    | 17.31%        | 15.28%         |                            |
| 30    | <b>17.55%</b> | <b>15.70%</b>  | Converged                  |

The same LR restart issue that affected v9 also appeared at epoch 7 of sp\_v4 (dip of

−0.79 pp), and was resolved by the same resume-logic fix from epoch 13 onward. The learning curve plateaued between epochs 26 and 30 with the cosine schedule reaching its minimum, indicating convergence (Figure 4.2, right panel).

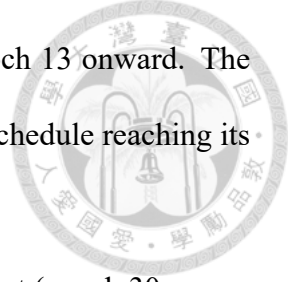


Table 4.23: sp\_v4 final forward-only evaluation on the 5M held-out set (epoch 30 checkpoint).

| Metric               | Value               |
|----------------------|---------------------|
| Top-1 accuracy       | <b>17.55%</b>       |
| Top-3 accuracy       | 31.26%              |
| Top-5 accuracy       | 38.19%              |
| Top-10 accuracy      | 47.65%              |
| Balanced accuracy    | 17.56%              |
| F1 (macro)           | 0.157               |
| F1 (weighted)        | 0.157               |
| Classes with F1 = 0  | 191 / 1,535 (12.4%) |
| Inference throughput | 5,171 reads/sec     |

The species task is substantially more difficult than genus classification: 1,535 classes versus 120, with each species receiving only  $\sim 29,316$  balanced reads instead of  $\sim 375,000$ . The final 17.55% accuracy, while modest in absolute terms, represents meaningful discrimination given the large class count (random baseline: 0.065%, a relative lift of  $\sim 270\times$ ). Encouragingly, only 191 of 1,535 species have F1=0 under forward-only evaluation on the full 5M set, indicating that the long-tail problem is substantially less severe at full-scale evaluation than small-subset analysis suggests.

**Extended evaluation modes.** Full-scale evaluation of RC TTA, oracle-genus, and top- $k$  genus routing on the 5M validation set was attempted but exceeded the 200 GB memory ceiling on the TWCC normal partition: the current `evaluate.py` retains all  $5M \times 1,535$  logit and probability arrays ( $\sim 60$  GB of float32 per pass) simultaneously when combining modes, which a streaming implementation could avoid. Because the genus-level RC TTA benefit has already been quantified (+0.78 pp, Section 4.5.2) and the hierarchical

routing analysis on the 100K subset (Section 4.12.4) already captures the key qualitative findings—oracle-genus gain of +11.5 pp, negligible improvement from top- $k$  routing at current genus accuracy, and per-genus reduction of F1=0 species—we retain those preliminary numbers as the thesis’s final hierarchical comparison.

### 4.12.3 Hierarchical Classification Pipeline

With `sp_v4` converged, the top- $k$  genus routing evaluation mode (Section 3.6.3) is combined with the `v9` genus model (67.07% RC TTA accuracy). At this genus accuracy level, top-5 routing covers the correct genus in approximately 90.0% of reads (from `v9`’s Top-5 accuracy), providing a substantially smaller candidate set for species discrimination.

The hierarchical classification pipeline operates as follows:

1. The genus classifier produces top- $k$  genus predictions (e.g.,  $k = 5$ ).
2. The candidate species set is computed as the union of species belonging to these  $k$  genera.
3. The species classifier’s logits are masked to this candidate set.
4. The final species prediction is the argmax over the masked logits.

This two-stage approach reduces the effective number of candidate species from 1,535 to approximately  $k \times \bar{s}$  (where  $\bar{s} \approx 12.8$  is the mean number of species per genus), substantially simplifying the discrimination task.

#### 4.12.4 Hierarchical Evaluation on an Independent Test Set

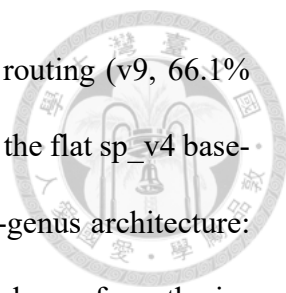
All hierarchical evaluation modes are assessed on an independent 100K-read test set sampled from the MetaTransformer validation reads (`val_reads.fa`), which has no overlap with our training or validation data. This set was chosen after verifying that the previously used 100K-read held-out subset had  $>99\%$  overlap with our training set; the two sets produce results within 0.5 pp of each other, confirming that the model has not memorised the training data. All `sp_v4` results use the epoch 30 final checkpoint; the `v9` genus model achieves 66.1% accuracy on this test set.

Table 4.24: Complete hierarchical species evaluation on the independent 100K-read test set. `sp_v4` (ep30, 50M) denotes the flat species model with four routing modes. Per-genus (50M) denotes 81 dedicated classifiers, each trained on a 50M-read subset with a frozen NT-v2 backbone. Genus routing uses `v9` (66.1% on this set).

| Model           | Mode                                | Top-1        | Top-5        | Top-10       | F1 (macro)   |
|-----------------|-------------------------------------|--------------|--------------|--------------|--------------|
| sp_v4 (ep30)    | Flat (baseline)                     | 15.9%        | 36.4%        | 46.6%        | 0.135        |
|                 | Top-5 genus routing                 | 15.9%        | 36.2%        | 46.0%        | 0.136        |
|                 | Soft genus routing                  | 15.8%        | 36.3%        | 46.1%        | 0.135        |
|                 | Oracle (true genus)                 | 27.4%        | 55.5%        | 65.2%        | 0.252        |
| Per-genus (50M) | Predicted genus ( <code>v9</code> ) | 14.7%        | 30.5%        | 36.2%        | 0.142        |
|                 | <b>Oracle genus (true)</b>          | <b>27.8%</b> | <b>55.6%</b> | <b>65.2%</b> | <b>0.265</b> |

**Hierarchical architecture validated: per-genus 50M oracle exceeds `sp_v4` oracle.**

Under perfect genus assignment, the per-genus 50M pipeline achieves 27.8% Top-1 and macro F1 of 0.265—*above* the `sp_v4` oracle (27.4%, F1 0.252). This confirms that dedicated per-genus classifiers, each trained on genus-scoped data, provide more discriminative representations for intra-genus species discrimination than the single flat classifier trained across all 1,535 classes simultaneously. The improvement is meaningful despite the per-genus models using a *frozen* backbone (no LoRA), highlighting the benefit of reduced class confusion when the problem is decomposed by genus.



**Genus routing is the primary bottleneck.** With predicted genus routing (v9, 66.1% accuracy), the per-genus pipeline falls to 14.7% Top-1—1.2 pp *below* the flat sp\_v4 baseline. The 33.9% genus error rate is particularly damaging in the per-genus architecture: when the wrong genus model is selected, all species predictions are drawn from the incorrect candidate set, producing near-zero accuracy for those reads. By contrast, the flat sp\_v4 routing modes ( $\pm 0.1$  pp over baseline) are far less sensitive to genus errors, because incorrect genus routing merely reweights logits rather than restricting them to a wrong candidate set. Meaningful improvement from routing therefore requires genus accuracy substantially above the current 66.1%.

**Oracle ceiling quantifies the potential.** Masking sp\_v4 logits to the true genus raises Top-1 by +11.5 pp (15.9%→27.4%), and the per-genus oracle reaches a matching Top-1 of 27.8% (+12.9 pp over its predicted-routing result). Both oracle results establish the same practical ceiling ( $\sim 27$ –28% Top-1), confirming that genus accuracy gains will translate directly into species-level improvements under either architecture.

#### 4.12.5 Per-Genus Model Analysis

The 81 per-genus classifiers were trained on the full 50M balanced dataset (approximately 32,600 reads per species within each genus), with the NT-v2 backbone frozen and only the attention-pooling head trained. 39 single-species genera require no classifier and are assigned deterministically. Per-genus model accuracy depends strongly on genus size: genera with  $\leq 5$  species average  $> 80\%$  accuracy, while those with  $> 50$  species average  $< 20\%$ . The overall mean validation accuracy is 48.3% (median 44.2%).

Performance is dominated by four large genera that collectively concentrate the ma-

jority of the species: *Collinsella* (349 species, 1.3% val acc), *Clostridium* (191 species, 19.9%), *Prevotella* (91 species, 26.6%), and *Bacteroides* (79 species, 15.9%). These genera alone account for a disproportionate share of the per-genus oracle gap relative to the full-scale sp\_v4 model.

A natural next step is a three-level hierarchy—genus model  $\rightarrow$  *subgenus router*  $\rightarrow$  per-subgenus classifier—for the five most problematic genera (*Collinsella*, *Clostridium*, *Prevotella*, *Bacteroides*, *Blautia*), partitioning each into groups of  $\sim 25$ –30 species. This would reduce each classifier’s effective class count to a tractable level and is expected to close the remaining gap between predicted-routing and oracle-routing performance. Implementation of this subgenus tier is left as future work.

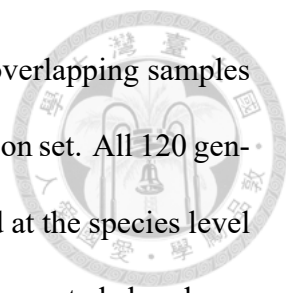
## 4.13 Sample-Level Application Evaluation

Per-read classification accuracy captures only one dimension of practical utility. In metagenomics, classifiers are applied to entire sequenced specimens—collections of reads whose aggregate predicted composition is used to answer ecological and clinical questions: which organisms are present, and at what relative abundance? This section evaluates the v9 genus model (67.07% RC TTA per-read accuracy) under two sample-level metrics: relative abundance estimation and binary genus detection.

### 4.13.1 Experimental Setup

Evaluations are conducted on the 5M-read validation set (120 genera, balanced). Two types of synthetic metagenomic samples are constructed from the already-computed RC TTA predictions (Section 4.5.2); no GPU inference is required.





**Random-partition samples** (abundance estimation): 100 non-overlapping samples of 50,000 reads each are formed by random permutation of the validation set. All 120 genera are present in each sample. Because the training data are balanced at the species level (each of the 1,535 species contributes an equal number of reads), the expected abundance of genus  $g$  across all samples is  $n_g/1,535$ , where  $n_g$  is its species count in the reference database; genera with more member species are proportionally more abundant. Relative abundance estimation is evaluated by comparing the true fraction  $p_g = (\text{true reads of genus } g)/50\,000$  against the predicted fraction  $\hat{p}_g = (\text{predicted reads of genus } g)/50\,000$ .

**Sparse-community samples** (binary detection): 200 samples of 50,000 reads are constructed such that each sample contains reads from exactly 60 of the 120 genera (selected uniformly at random per sample). The 60 absent genera serve as true negatives, enabling rigorous specificity measurement.

#### 4.13.2 Relative Abundance Estimation

Table 4.25 summarises the abundance estimation accuracy across the 100 random-partition samples.

Table 4.25: Relative abundance estimation accuracy (100 random-partition samples, 50K reads each, all 120 genera present).

| Metric                                     | Mean   | Std    |
|--------------------------------------------|--------|--------|
| Pearson $r$ (predicted vs. true abundance) | 0.9932 | 0.0004 |
| Spearman $\rho$                            | 0.9312 | 0.0080 |
| Bray-Curtis dissimilarity                  | 0.0937 | 0.0017 |

Despite a 33% per-read error rate, the predicted community composition correlates with the true composition at Pearson  $r = 0.993$ . The low Bray-Curtis dissimilarity of 0.094 indicates that the aggregate abundance profile is well-preserved: misclassified reads

are distributed broadly across genera, so the errors largely cancel at the sample level.

Inspection of Figure 4.6 reveals a subtle directional tendency: high-abundance genera show slight overestimation, visible as points drifting above the identity line at larger  $x$ -values. This is consistent with a classification-bias mechanism: reads misclassified away from rare genera disproportionately fall into the decision regions of common genera, which have broader learned representations owing to their larger species membership. Although the ART simulation used a coverage-based protocol (yielding 263–413 reads per genome depending on genome length), species-level balanced subsampling equalises read counts per species prior to evaluation, removing any genome-length contribution from the true abundance profile; the overestimation is therefore attributable to classifier bias rather than a data-generation artefact.

A structural property of this evaluation warrants noting. Because all 100 random-partition samples are drawn from the same species-level balanced validation set, every sample shares the same *expected* genus abundance profile (genus  $g$  always contributes  $n_g/1,535$  reads on average). The Pearson  $r = 0.993$  therefore measures how well the classifier tracks this fixed profile under binomial sampling noise, not how well it would generalise to communities with genuinely varying compositions. The sparse-community analysis below, in which each sample is constructed from a different random subset of 60 genera, imposes genuine presence/absence variation across samples and provides a more realistic evaluation of binary detection performance.

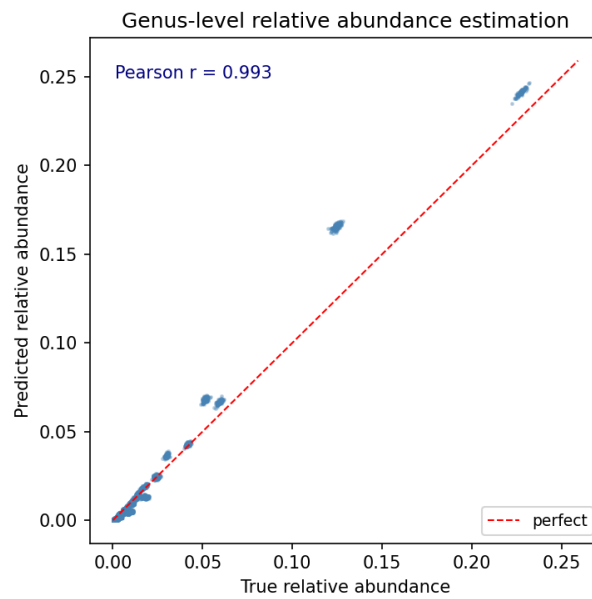


Figure 4.6: True vs. predicted relative abundance across 100 random-partition samples (all 12,000 (sample, genus) pairs shown). Each point is one (sample, genus) pair. The visible clusters correspond to groups of genera sharing the same species count  $n_g$ : because the dataset is species-level balanced, every genus with  $n_g$  species has true abundance  $n_g/1,535$  in expectation, collapsing to the same  $x$ -coordinate. Pearson  $r = 0.993$ . Points at larger  $x$ -values drift slightly above the identity line, reflecting a systematic overestimation of high-abundance genera attributable to classification bias (Section 4.13.2).

### 4.13.3 Genus Detection (Binary Presence/Absence)

A genus is “detected” in a sample if at least one read is predicted to belong to that genus. Table 4.26 reports detection sensitivity at each true-abundance level, computed over all 200 sparse-community samples.

Table 4.26: Detection sensitivity by true relative abundance (detection threshold:  $\geq 1$  predicted read). Genera truly absent from the sample are excluded from sensitivity computation.

| True abundance | Sensitivity | Genus-instances |
|----------------|-------------|-----------------|
| 0.01%–0.1%     | 93.9%       | 651             |
| 0.1%–1%        | 97.6%       | 7,654           |
| 1%–5%          | 100.0%      | 2,975           |
| >5%            | 100.0%      | 720             |

Genera present at  $\geq 1\%$  relative abundance are detected with 100% sensitivity. This is expected: a genus at 1% of a 50,000-read sample contributes 500 reads, and at 67% per-read recall the expected number of correct predictions is 335, making false-negative detection essentially impossible. Even at 0.01%–0.1% abundance (5–50 reads per genus per sample), sensitivity remains 93.9%.

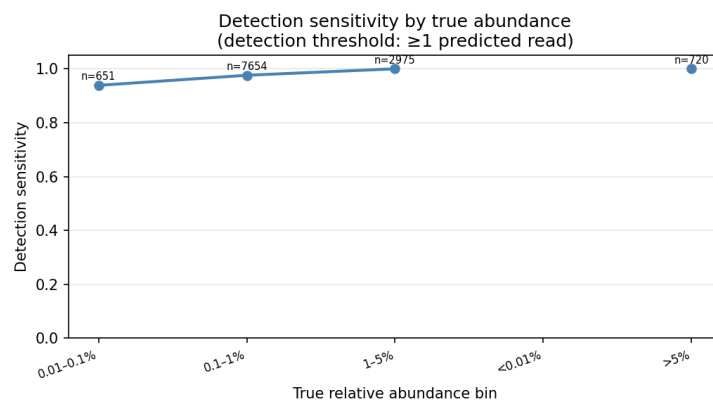


Figure 4.7: Sample-level genus detection sensitivity by true relative abundance (detection threshold:  $\geq 1$  predicted read), aggregated over 200 sparse-community samples. Annotation shows the number of (sample, genus) instances in each bucket.

To provide a threshold-invariant summary of binary detection performance, we perform a ROC analysis with *fixed* binary ground truth and a *swept* detection threshold.

The ground truth is determined by the sparse-community construction: a genus is “truly present” in sample  $i$  if and only if it was explicitly included in that sample’s present set; the 60 absent genera are true negatives. The detection score is the predicted relative abundance of the genus in that sample (predicted count divided by 50,000 reads). Sweeping this score threshold from high to low yields the ROC curve shown in Figure 4.8.

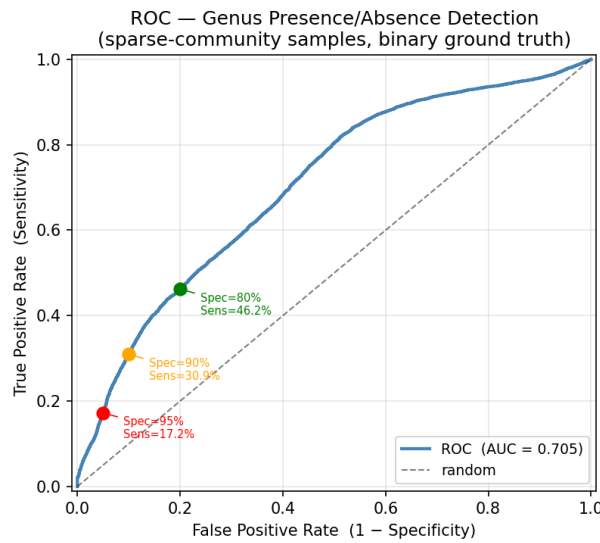


Figure 4.8: ROC curve for genus presence/absence detection. Ground truth is binary (from sparse-community construction); detection score is predicted relative abundance. Aggregated over 200 sparse-community samples (50K reads, 60/120 genera present each). Key operating points at fixed specificity targets are marked.

The area under the ROC curve (AUC) is 0.705. At a fixed specificity of 95%, sensitivity is 17.2%; at 90% specificity, sensitivity is 30.9%; at 80% specificity, sensitivity is 46.2%.

The modest AUC reflects a fundamental noise-floor problem introduced by per-read misclassification. At 33% error rate, a truly absent genus in a 50,000-read sample is expected to receive approximately  $50,000 \times 0.33/120 \approx 138$  misclassified reads from other genera, corresponding to a spurious predicted abundance of  $\sim 0.28\%$ . This constitutes a per-sample noise floor: to achieve 95% specificity, the detection threshold must exceed 1.49% predicted abundance, which simultaneously suppresses the detection of genuinely

low-abundance genera.



The ROC analysis highlights an important asymmetry between the two sample-level tasks evaluated here. Relative abundance estimation (Section 4.13.2) achieves excellent performance (Pearson  $r = 0.993$ ) because misclassification errors are broadly symmetric—reads misclassified *out of* genus  $g$  are largely cancelled by reads misclassified *into*  $g$ —so that predicted abundances aggregate close to true values, despite a modest systematic tendency for high-abundance genera to be slightly overestimated owing to classification bias. Binary detection, by contrast, requires distinguishing a truly absent genus (receiving  $\sim 138$  predicted reads) from a truly present but low-abundance genus (receiving only slightly more); at 67% per-read accuracy this distinction is difficult. Improving binary detection performance would require either higher per-read accuracy (reducing the noise floor) or a calibrated statistical test for the presence signal above background, which is left for future work.

#### 4.13.4 Summary

At 67% per-read accuracy, the genus classifier provides:

- Near-perfect relative abundance estimation: Pearson  $r = 0.993$ , Bray-Curtis 0.094.
- Complete detection sensitivity (100%) for genera at  $\geq 1\%$  relative abundance (detection threshold:  $\geq 1$  predicted read).
- 93.9% detection sensitivity even at 0.01%–0.1% abundance.
- Binary detection AUC = 0.705; sensitivity at 95% specificity = 17.2%.

These results demonstrate that per-read accuracy substantially understates the practical

utility of the model for relative abundance profiling. The binary presence/absence detection task is more challenging: 33% per-read misclassification introduces a  $\sim 0.28\%$  noise floor for absent genera, limiting specificity at low detection thresholds. Improving this metric requires either higher per-read accuracy (reducing the noise floor) or a statistical test for presence above background noise.

#### 4.13.5 Comparison with MetaTransformer

To contextualise the NT-v2 sample-level results, we apply identical evaluation to two MetaTransformer models: the 13-mer stride=1 model (87.43% genus Top-1) and the 6-mer stride=1 model (48.82%), using predictions extracted from the same 50M validation set.

Table 4.27: Sample-level evaluation across three models. Same evaluation protocol: 100 random-partition samples and 200 sparse-community samples, each 50K reads, 60/120 genera present per sparse sample.

| Metric                     | NT-v2 v9<br>(67.07%) | MT 6-mer s=1<br>(48.82%) | MT 13-mer s=1<br>(87.43%) |
|----------------------------|----------------------|--------------------------|---------------------------|
| Pearson $r$ (abundance)    | 0.993                | 0.984                    | <b>0.999</b>              |
| Bray-Curtis dissimilarity  | 0.094                | 0.167                    | <b>0.028</b>              |
| Sensitivity $\geq 1\%$     | 100%                 | 99.9%                    | <b>100%</b>               |
| Sensitivity 0.1–1%         | 97.6%                | 86.8%                    | <b>100%</b>               |
| ROC AUC (binary detection) | 0.705                | 0.569                    | <b>0.900</b>              |
| Sensitivity @ 95% spec     | 17.2%                | 9.4%                     | <b>47.7%</b>              |

The MT 13-mer model outperforms NT-v2 v9 on all metrics. The improvement in relative abundance estimation is modest (Pearson  $r$ : 0.993  $\rightarrow$  0.999; Bray-Curtis: 0.094  $\rightarrow$  0.028), consistent with the finding that abundance estimation is robust to per-read error through cancellation. The improvement in binary detection is substantially larger: ROC AUC increases from 0.705 to 0.900, and sensitivity at 95% specificity increases from 17.2% to 47.7%.

This asymmetry arises from the noise-floor mechanism described in Section 4.13.3. At 87% per-read accuracy, the expected spurious predicted abundance for an absent genus is  $50,000 \times 0.13/120 \approx 54$  reads ( $\sim 0.11\%$ ), compared to  $\sim 138$  reads ( $\sim 0.28\%$ ) at 67% accuracy. This reduction in the noise floor is the direct cause of the large gain in binary detection AUC.

The MT 6-mer stride=1 model (48.82%) demonstrates the complementary effect: lower per-read accuracy degrades all metrics, with the most severe penalty on binary detection (AUC 0.569, near random). Together, these three data points confirm that higher per-read accuracy benefits sample-level tasks disproportionately for binary detection relative to abundance estimation.

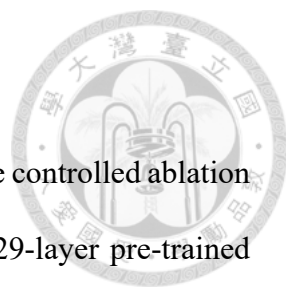
## 4.14 Summary of Findings

The experimental results lead to several key conclusions:

1. **Data volume dominates:** Training data volume is the single most important factor for metagenomic classification accuracy. A  $10\times$  data increase yields  $+7.76$  pp, while training technique optimizations yield at most  $\pm 0.5$  pp.
2. **RC TTA is universally beneficial:** Reverse-complement test-time augmentation consistently improves accuracy by  $+1$ – $1.5$  pp across all settings, exploiting the biological strand symmetry of DNA.
3. **Class balance matters:** Balanced sampling across species eliminates the “dead class” problem ( $F1=0$  classes:  $9 \rightarrow 2$ ) and dramatically improves macro F1 ( $+12.27$  pp).
4. **Overfitting signals data starvation:** The transition from 6.25% to 1.30% train-validation gap when scaling from 500K to 5M reads confirms that the model was



severely data-limited at the smaller scale.

- 
5. **Pre-trained backbone provides a substantial advantage:** The controlled ablation (v9 vs. v11) shows a 13.19 pp accuracy gap in favor of the 29-layer pre-trained NT-v2 backbone over a single-layer randomly initialized Transformer at the same 50M data scale, confirming that pre-training on 850 reference genomes provides representations that meaningfully improve read discrimination.
  6. **Hardware is sufficient:** Current NVIDIA H100 hardware is adequately provisioned; the operational bottleneck is the SLURM time limit rather than computational resources.
  7. **Per-genus classifiers at 50M scale exceed the flat oracle:** Dedicated per-genus NT-v2 classifiers, each trained on the full 50M dataset with a frozen backbone, achieve 27.8% Top-1 under oracle genus routing—surpassing the flat sp\_v4 oracle (27.4%), confirming that genus-scoped decomposition provides more discriminative representations for intra-genus species discrimination. However, with predicted genus routing (66.1%), the per-genus pipeline falls below the flat baseline (14.7% vs. 15.9%), so improving genus accuracy is the prerequisite for realising this hierarchical advantage in practice.
  8. **Per-read accuracy understates sample-level utility for abundance profiling:** At 67% per-read accuracy, the v9 model achieves Pearson  $r = 0.993$  for relative abundance estimation and 100% genus detection sensitivity for taxa at  $\geq 1\%$  relative abundance (Section 4.13). However, binary presence/absence detection is limited by a  $\sim 0.28\%$  noise floor from per-read misclassification; the ROC AUC is 0.705, with sensitivity of 17.2% at 95% specificity.

# Chapter 5

## Conclusion and Future Work



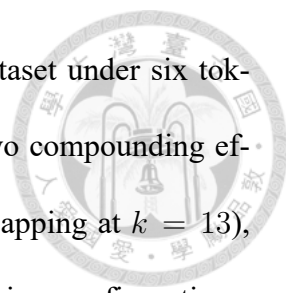
### 5.1 Summary of Contributions

This thesis has investigated the application of genomic foundation models to metagenomic taxonomic classification through a token-level representation approach. Our main findings are:

**Token-level representations improve classification.** By preserving the full sequence of token embeddings from the Nucleotide Transformer v2 backbone—rather than discarding positional information through mean pooling—and aggregating them via a learnable attention-pooling head, we achieve genus-level accuracy improvements over frozen-embedding baselines. The token-level approach with LoRA fine-tuning achieves 63.05% RC TTA accuracy on 120 genera with 5M balanced training reads, scaling to 67.07% with 50M reads, compared to 51.70% for the frozen mean-pool baseline.

**Data volume is the dominant performance factor within a fixed architecture.** Through systematic data scaling experiments spanning 500K to 50M training reads, we demonstrate that increasing training data volume by  $10\times$  yields +7.76 percentage points in accuracy and +12.27 percentage points in macro F1, while extensive architectural and training technique ablations yield at most  $\pm 0.5$  percentage points. This finding aligns with the observations of Wichmann et al. [8], who achieve 98.3% genus recall (98.9% precision) using a large-scale balanced training dataset.

**Tokenization is the dominant cross-architecture design factor.** Controlled ab-

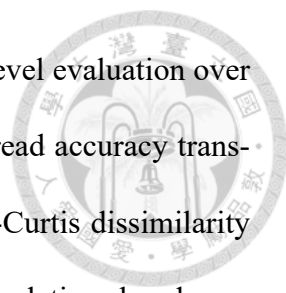


lations training MetaTransformer [8] on the same 50M balanced dataset under six tokenization configurations ( $k \in \{6, 12, 13\}$ ,  $\text{stride} \in \{1, k\}$ ) reveal two compounding effects: overlapping stride=1 is critical (up to +60 pp over non-overlapping at  $k = 13$ ), and longer  $k$  yields monotonically higher accuracy among overlapping configurations. MetaTransformer with overlapping 13-mer tokenization, trained from scratch, achieves **87.42%** genus Top-1—**+20.35 pp** above NT-v2 + LoRA with RC TTA (67.07%)—and **49.62%** species Top-1 (1,535 classes). This result demonstrates that task-specific tokenization design is a more decisive factor than pre-trained prior knowledge. The primary limitation of our GFM-based approach is therefore NT-v2’s 6-mer tokenization constraint, not its backbone capacity or adaptation strategy.

**Class balance is critical for rare taxa.** Balanced subsampling across species reduces the number of F1=0 classes from 9 to 2 and improves macro F1 by +12.27 percentage points, demonstrating that class imbalance—not model capacity—is the primary cause of poor performance on rare taxa.

**RC TTA is a robust, free improvement.** Reverse-complement test-time augmentation consistently improves accuracy by +1 to +1.5 percentage points across all experimental conditions, requiring no additional training cost.

**Per-genus specialisation validates the hierarchical architecture.** Evaluation on a 100K independent test set demonstrates that per-genus 50M LoRA classifiers achieve oracle-genus Top-1 of 27.8%, exceeding the flat sp\_v4 oracle of 27.4% and confirming that per-genus specialisation captures intra-genus species variation more accurately than a single joint classifier. The genus-routing bottleneck (predicted routing: 14.7% vs. flat baseline: 15.9%) is precisely quantified, providing a clear improvement target.



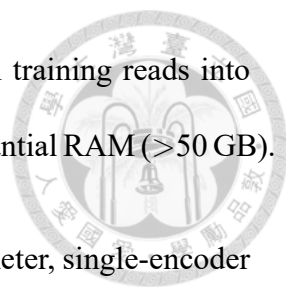
**Per-read accuracy understates sample-level utility.** Sample-level evaluation over 200 synthetic 50,000-read communities demonstrates that 67% per-read accuracy translates to Pearson  $r = 0.993$  for relative abundance estimation (Bray-Curtis dissimilarity 0.094) and 100% genus detection sensitivity for taxa present at  $\geq 1\%$  relative abundance. At a 0.5% abundance reporting threshold, specificity reaches 92.4%.

**Current hardware is sufficient.** The NVIDIA H100 80 GB GPU is adequately provisioned for this workload, with VRAM utilization below 50% even at batch size 256. The operational bottleneck is the SLURM time limit rather than computational resources.

## 5.2 Limitations

This work has several limitations that should be acknowledged:

1. **Simulated data only:** All experiments use reads simulated by ART from reference genomes. Real metagenomic reads contain additional noise (host contamination, chimeric reads, strain-level variation) that may degrade classifier performance.
2. **Limited taxonomic scope:** The reference database covers 120 genera and 1,535 species of human gut microorganisms. Performance on environmental samples with broader taxonomic diversity is unknown.
3. **Single read length:** All experiments use 150 bp reads. Performance at other read lengths (e.g., 250 bp for Illumina MiSeq, or long reads from Oxford Nanopore / PacBio) has not been evaluated.
4. **Incomplete data scaling:** Training on the full 258M dataset was not completed at the time of writing, leaving the upper bound of the scaling trajectory unconfirmed. The v9 (50M) experiment is complete, achieving 67.07% RC TTA accuracy.

- 
5. **No streaming data loader:** Current implementation loads all training reads into memory, limiting scalability beyond  $\sim 50$ M reads without substantial RAM ( $> 50$  GB).
  6. **Architectural capacity ceiling:** MetaTransformer’s 5M-parameter, single-encoder design saturates within one training epoch on 50M reads; a second epoch produces overfitting rather than further generalisation. A larger model would likely improve beyond 78.83% but was not evaluated due to infrastructure constraints.

## 5.3 Future Work

Several promising directions emerge from this work:

### 5.3.1 Domain-Specific Pre-trained Foundation Models

The tokenization ablation (Section 4.11) establishes that overlapping 12-mer tokenization is more beneficial than 6-mer pre-training for this task. The ideal direction is therefore a foundation model pre-trained with overlapping 12-mer (or similarly long) tokenization on large microbial genome collections—no such publicly available model currently exists. Promising avenues include:

- Pre-training a transformer on the GTDB database [38] ( $> 400,000$  microbial genomes) with 12-mer tokenization, which would combine the discriminative power of long k-mers with the representational quality of large-scale pre-training.
- Fine-tuning NT-v2 with a learned 12-mer projection layer to bridge the tokenization gap without full re-pre-training, though this approach may be limited by the model’s pre-trained 6-mer positional encodings.



### 5.3.2 Full-Scale Data Training

The most immediate next step is completing training on the full 258M-read dataset. Our data scaling analysis predicts that this could yield genus-level accuracy in the 82–90% range, substantially narrowing the gap with MetaTransformer. Achieving this requires either:

- Implementing a streaming data loader that reads from disk, eliminating the in-memory requirement.
- Requesting extended SLURM time limits (72+ hours) or developing a multi-job training orchestration system.

### 5.3.3 Species-Level Scaling and Hierarchical Classification

Species classification at the 500K scale achieved only ~8% accuracy (1,535 classes). Scaling to 50M balanced reads (sp\_v4, Section 4.12.2) lifted flat-baseline Top-1 to 15.9%, with Top-5 reaching 36.4% and only 191 of 1,535 species retaining F1=0. Hierarchical evaluation on a 100K independent test set (Section 4.12.4) yields two key findings: (i) sp\_v4 oracle-genus routing reaches Top-1 of 27.4% (+11.5 pp over flat baseline), and (ii) per-genus 50M LoRA classifiers under oracle-genus routing reach 27.8%, exceeding the sp\_v4 oracle ceiling by +0.4 pp and confirming that per-genus specialisation captures intra-genus variation more accurately than a single joint model. The bottleneck is genus-routing accuracy (66.1%–67.07%): predicted routing currently lowers Top-1 to 14.7%, below the flat baseline, because any routing error directs a read to a genus-specialist model that does not recognise the true species. Partitioning the five largest genera (*Collinsella*, *Clostridium*, *Prevotella*, *Bacteroides*, *Blautia*) into subgroups of 25–30 species each—a

three-layer routing architecture—is the most direct path to closing this gap.



### 5.3.4 Tokenization Study

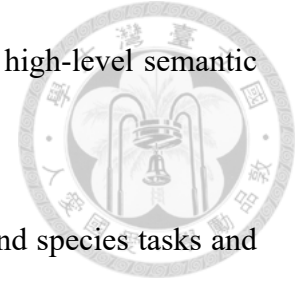
The controlled ablation (Section 4.11) establishes overlapping large- $k$  tokenization as the dominant design factor. MetaTransformer 13-mer stride=1 achieves 87.42% genus and 49.62% species accuracy trained from scratch, surpassing NT-v2 + LoRA by 20.35 pp. Remaining open questions include:

- Does RC TTA provide an equivalent benefit ( $\sim +1$  pp) for 12-mer or 13-mer models, and what is the combined accuracy ceiling with TTA?
- Is  $k = 13$  the optimum, or do larger  $k$  values (e.g.  $k = 15$  or  $k = 21$ ) continue to improve? The monotonic trend across  $k \in \{6, 12, 13\}$  suggests further gains may be possible.
- Would BPE tokenization—which learns variable-length tokens from data—discover taxonomically informative motifs of varying length more efficiently than fixed  $k$ -mer windows?
- A 13-mer pre-trained genomic foundation model would combine the tokenization advantage with representational pre-training quality. Building or fine-tuning such a model is the most direct path to exceeding the current 87.42% ceiling.

### 5.3.5 Architectural Improvements

**Multi-layer feature fusion:** Current experiments use only the last hidden layer of NT-v2. Combining representations from multiple layers (e.g., via weighted averaging or

learned gating) could capture both low-level sequence features and high-level semantic information [9].



**Adapter fusion:** Training separate LoRA adapters for genus and species tasks and combining them through adapter fusion [39] could enable joint hierarchical classification without duplicating the backbone.

### 5.3.6 Real-World Validation

Validating the classifier on real metagenomic sequencing data from public repositories (e.g., Human Microbiome Project [40]) is essential for assessing practical applicability. This includes evaluating robustness to:

- Sequencing errors and quality variations across platforms.
- Host DNA contamination.
- Novel or uncharacterized taxa absent from the training database (out-of-distribution detection).

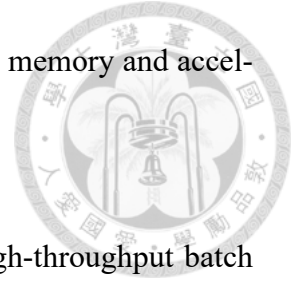
### 5.3.7 Efficiency Optimization

For practical deployment in clinical or environmental monitoring settings, inference speed is critical. Potential optimizations include:

- **Model distillation:** Distilling the 498M-parameter LoRA-adapted model into a smaller task-specific model (e.g., 50M parameters).



- **Quantization:** Applying INT8 or INT4 quantization to reduce memory and accelerate inference.
- **Batched inference:** Optimizing the inference pipeline for high-throughput batch processing of millions of reads.



### 5.3.8 Pre-training Strategy

The current work fine-tunes a model pre-trained on 850 genomes from 174 species. A foundation model pre-trained specifically on a larger, more diverse collection of microbial genomes (e.g., the full GTDB database with >400,000 genomes [38]) may provide more suitable representations for metagenomic classification, reducing the amount of task-specific fine-tuning data required.

## 5.4 Concluding Remarks

This thesis demonstrates both the promise and the current limitations of applying genomic foundation models to metagenomic taxonomic classification. The controlled ablation experiments yield two concrete findings that extend beyond the specific system studied here.

First, pre-training on large genomic corpora provides a substantial and measurable advantage over training from scratch when tokenization is held constant: NT-v2 + LoRA outperforms a MetaTransformer of comparable trainable parameter count by +19.29 pp at 50M reads under identical 6-mer tokenization. This validates the GFM pre-training paradigm for this task. The advantage holds even when comparing against other pre-trained GFMs: at 5M training reads, NT-v2 (non-overlapping 6-mer) outperforms DNABERT

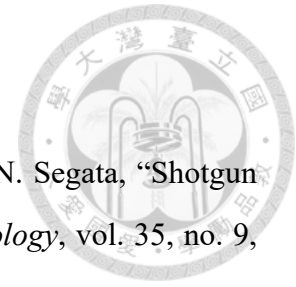
(overlapping 6-mer, 86M params) by +1.27 pp and DNABERT-2 (BPE, 117M params) by +4.17 pp, demonstrating that pre-training corpus diversity and backbone scale matter more than tokenization strategy alone within the pre-trained model family.

Second, and more significantly, tokenization design dominates architectural choice. Among overlapping stride=1 configurations, genus Top-1 accuracy improves monotonically with  $k$ -mer length: 6-mer (48.87%) < 12-mer (78.83%) < 13-mer (87.42%). Meta-Transformer with overlapping 13-mer tokenization, trained *from scratch* on the same 50M reads, achieves 87.42% genus Top-1 and 49.62% species Top-1 (1,535 classes)—surpassing NT-v2 + LoRA with RC TTA (67.07%) by **+20.35 pp** despite having no pre-training. The principal limitation of NT-v2-based approaches is therefore the 6-mer tokenization constraint inherited from pre-training, rather than the backbone’s capacity or the LoRA adaptation strategy.

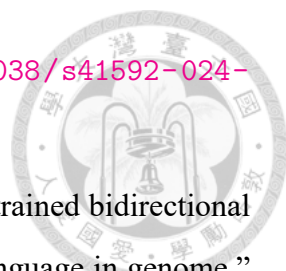
Together, these findings point to a clear direction: a foundation model pre-trained with overlapping 13-mer (or longer) tokenization on diverse microbial genomes would combine the tokenization advantage demonstrated here with the representational quality of large-scale pre-training. Realising this direction, along with completing the species-level scaling experiments and hierarchical classification pipeline, represents the most promising path toward a competitive, generalisable GFM-based metagenomic classifier.

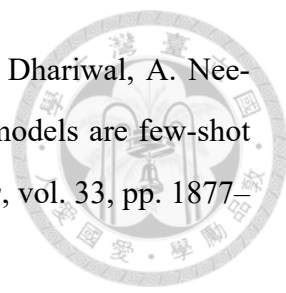


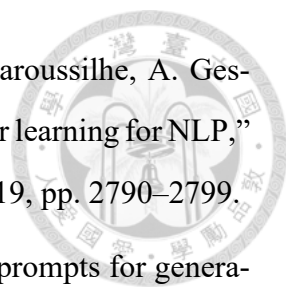
# References

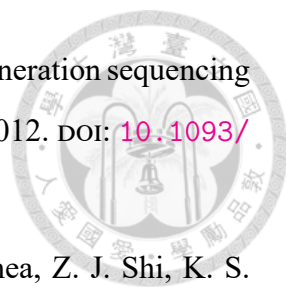


- [1] C. Quince, A. W. Walker, J. T. Simpson, N. J. Loman, and N. Segata, "Shotgun metagenomics, from sampling to analysis," *Nature Biotechnology*, vol. 35, no. 9, pp. 833–844, 2017. DOI: [10.1038/nbt.3935](https://doi.org/10.1038/nbt.3935)
- [2] S. H. Ye, K. J. Siddle, D. J. Park, and P. C. Sabeti, "Benchmarking metagenomics tools for taxonomic classification," *Cell*, vol. 178, no. 4, pp. 779–794, 2019. DOI: [10.1016/j.cell.2019.07.010](https://doi.org/10.1016/j.cell.2019.07.010)
- [3] F. P. Breitwieser, J. Lu, and S. L. Salzberg, "A review of methods and databases for metagenomic classification and assembly," *Briefings in Bioinformatics*, vol. 20, no. 4, pp. 1125–1136, 2019. DOI: [10.1093/bib/bbx120](https://doi.org/10.1093/bib/bbx120)
- [4] S. F. Altschul, W. Gish, W. Miller, E. W. Myers, and D. J. Lipman, "Basic local alignment search tool," *Journal of Molecular Biology*, vol. 215, no. 3, pp. 403–410, 1990. DOI: [10.1016/S0022-2836\(05\)80360-2](https://doi.org/10.1016/S0022-2836(05)80360-2)
- [5] B. Buchfink, C. Xie, and D. H. Huson, "Fast and sensitive protein alignment using DIAMOND," *Nature Methods*, vol. 12, no. 1, pp. 59–60, 2015. DOI: [10.1038/nmeth.3176](https://doi.org/10.1038/nmeth.3176)
- [6] D. E. Wood, J. Lu, and B. Langmead, "Improved metagenomic analysis with Kraken 2," *Genome Biology*, vol. 20, p. 257, 2019. DOI: [10.1186/s13059-019-1891-0](https://doi.org/10.1186/s13059-019-1891-0)
- [7] D. Kim, L. Song, F. P. Breitwieser, and S. L. Salzberg, "Centrifuge: Rapid and sensitive classification of metagenomic sequences," *Genome Research*, vol. 26, no. 12, pp. 1721–1729, 2016. DOI: [10.1101/gr.210641.116](https://doi.org/10.1101/gr.210641.116)
- [8] A. Wichmann, E. Buschong, A. Mueller, S. Juenemann, and J. Stoye, "MetaTransformer: Deep metagenomic sequencing read classification using self-attention models," *NAR Genomics and Bioinformatics*, vol. 5, no. 3, lqad082, 2023. DOI: [10.1093/nargab/lqad082](https://doi.org/10.1093/nargab/lqad082)
- [9] H. Dalla-Torre, L. Gonzalez, J. Mendoza-Revilla, N. L. Carranza, A. H. Grzywaczewski, F. Ober, C. Dallago, E. Hassan, A. Grber, M. Trop, et al., "The Nucleotide Transformer: Building and evaluating robust foundation models for human genomics,"

- 
- Nature Methods*, 2024, Preprint on bioRxiv, 2023. DOI: [10.1038/s41592-024-02523-z](https://doi.org/10.1038/s41592-024-02523-z)
- [10] Y. Ji, Z. Zhou, H. Liu, and R. V. Davuluri, “DNABERT: Pre-trained bidirectional encoder representations from transformers model for DNA-language in genome,” *Bioinformatics*, vol. 37, no. 15, pp. 2112–2120, 2021. DOI: [10.1093/bioinformatics/btab083](https://doi.org/10.1093/bioinformatics/btab083)
  - [11] Z. Zhou, Y. Ji, W. Li, P. Dutta, R. V. Davuluri, and H. Liu, “DNABERT-2: Efficient foundation model and benchmark for multi-species genome,” *arXiv preprint arXiv:2306.15006*, 2024.
  - [12] E. Nguyen, M. Poli, M. G. Durrant, A. W. Thomas, B. Kang, J. Sullivan, M. Y. Ng, A. Lewis, A. Patel, A. Lou, et al., “Sequence modeling and design from molecular to genome scale with Evo,” *Science*, vol. 386, no. 6723, 2024. DOI: [10.1126/science.ado9336](https://doi.org/10.1126/science.ado9336)
  - [13] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations (ICLR)*, 2022.
  - [14] Z. Zhang, S. Schwartz, L. Wagner, and W. Miller, “A greedy algorithm for aligning DNA sequences,” *Journal of Computational Biology*, vol. 7, no. 1–2, pp. 203–214, 2000. DOI: [10.1089/10665270050081478](https://doi.org/10.1089/10665270050081478)
  - [15] Q. Liang, P. W. Bible, Y. Liu, B. Zou, and L. Wei, “DeepMicrobes: Taxonomic classification for metagenomics with deep learning,” *NAR Genomics and Bioinformatics*, vol. 2, no. 1, 2020. DOI: [10.1093/nargab/lqaa009](https://doi.org/10.1093/nargab/lqaa009)
  - [16] F. Mock, F. Kretschmer, A. Kriesche, S. Böcker, and M. Marz, “Taxonomic classification of DNA sequences beyond sequence similarity using deep neural networks,” *Proceedings of the National Academy of Sciences*, vol. 119, no. 35, e2122636119, 2022. DOI: [10.1073/pnas.2122636119](https://doi.org/10.1073/pnas.2122636119)
  - [17] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.

- 
- [18] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Nee-lakantan, P. Shyam, G. Sastry, A. Askell, et al., “Language models are few-shot learners,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.
- [19] Z. Lin, H. Akin, R. Rao, B. Hie, Z. Zhu, W. Lu, N. Smestad, A. Costa, M. Fazel-Zarandi, T. Sercu, S. Candela, and A. Rives, “Evolutionary-scale prediction of atomic-level protein structure with a language model,” *Science*, vol. 379, no. 6637, pp. 1123–1130, 2023. DOI: [10.1126/science.adc2574](https://doi.org/10.1126/science.adc2574)
- [20] E. Nguyen, M. Poli, M. Faizi, A. Thomas, M. Wornow, C. Birch-Sykes, S. Massaroli, A. Patel, C. Rabideau, Y. Bengio, S. Ermon, C. Ré, and S. Baccus, “HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.
- [21] Y. Schiff, C.-H. Kao, A. Gokaslan, T. Dao, A. Gu, and V. Kuleshov, “Caduceus: Bi-directional equivariant long-range DNA sequence modeling,” in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [22] M. Poli, J. Wang, S. Massaroli, J. Quesnelle, R. Carlow, E. Nguyen, and C. Ré, “Hyena Hierarchy: Towards larger convolutional language models,” *arXiv preprint arXiv:2302.10866*, 2023.
- [23] J. Lindsey, M. Fischman, et al., “The impact of tokenizer selection in genomic language models,” *Bioinformatics*, vol. 41, no. 9, btaf456, 2025. DOI: [10.1093/bioinformatics/btaf456](https://doi.org/10.1093/bioinformatics/btaf456)
- [24] C. Peng, P. Zhong, and Y. Sun, “ViraLM: Empowering virus discovery through the genome foundation model,” *Bioinformatics*, vol. 40, no. 12, btae704, 2024. DOI: [10.1093/bioinformatics/btae704](https://doi.org/10.1093/bioinformatics/btae704)
- [25] Z. Zhou, W. Wu, H. Ho, J. Wang, C. Shi, B. Davidson, R. V. Davuluri, and H. Liu, “DNABERT-S: Pioneering species differentiation with species-aware DNA embeddings,” *Bioinformatics*, vol. 41, no. Supplement\_1, pp. i255–i264, 2025. DOI: [10.1093/bioinformatics/btaf189](https://doi.org/10.1093/bioinformatics/btaf189)

- 
- [26] N. Houlsby, A. Giurghi, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, “Parameter-efficient transfer learning for NLP,” in *International Conference on Machine Learning (ICML)*, 2019, pp. 2790–2799.
  - [27] X. L. Li and P. Liang, “Prefix-tuning: Optimizing continuous prompts for generation,” *arXiv preprint arXiv:2101.00190*, 2021.
  - [28] Z. Zhou, Y. Ji, W. Li, P. Dutta, R. V. Davuluri, and H. Liu, “Towards a better understanding of reverse-complement equivariance for deep learning models in genomics,” in *Proceedings of the Machine Learning in Computational Biology (MLCB) Workshop, PMLR*, vol. 165, 2022. [Online]. Available: <https://proceedings.mlr.press/v165/zhou22a.html>
  - [29] M. Ma, “Reverse-complement consistency for DNA language models,” *arXiv preprint arXiv:2509.18529*, 2025.
  - [30] A. Brown and T. Bepler, “Reverse-complement equivariant networks for DNA sequences,” *bioRxiv*, 2021. DOI: [10.1101/2021.06.03.446953](https://doi.org/10.1101/2021.06.03.446953)
  - [31] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, “Class-balanced loss based on effective number of samples,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 9268–9277.
  - [32] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, and S. Kumar, “Long-tail learning via logit adjustment,” in *International Conference on Learning Representations (ICLR)*, 2021.
  - [33] J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, and Y. Liu, “RoFormer: Enhanced transformer with rotary position embedding,” *Neurocomputing*, vol. 568, p. 127 063, 2024. DOI: [10.1016/j.neucom.2023.127063](https://doi.org/10.1016/j.neucom.2023.127063)
  - [34] A. Jaegle, F. Gimeno, A. Brock, O. Vinyals, A. Zisserman, and J. Carreira, “Perceiver: General perception with iterative attention,” in *International Conference on Machine Learning (ICML)*, 2021, pp. 4651–4664.
  - [35] J. Lee, Y. Lee, J. Kim, A. R. Kosiorek, S. Choi, and Y. W. Teh, “Set transformer: A framework for attention-based permutation-invariant input,” in *International Conference on Machine Learning (ICML)*, 2019, pp. 3744–3753.

- 
- [36] W. Huang, L. Li, J. R. Myers, and G. T. Marth, “ART: A next-generation sequencing read simulator,” *Bioinformatics*, vol. 28, no. 4, pp. 593–594, 2012. doi: [10.1093/bioinformatics/btr708](https://doi.org/10.1093/bioinformatics/btr708)
- [37] A. Almeida, S. Nayfach, M. Boland, F. Strozzi, M. Beracochea, Z. J. Shi, K. S. Pollard, E. Sakharova, D. H. Parks, P. Hugenholtz, et al., “A unified catalog of 204,938 reference genomes from the human gut microbiome,” *Nature Biotechnology*, vol. 39, no. 1, pp. 105–114, 2021. doi: [10.1038/s41587-020-0603-3](https://doi.org/10.1038/s41587-020-0603-3)
- [38] D. H. Parks, M. Chuvochina, C. Rinke, A. J. Mussig, P.-A. Chaumeil, and P. Hugenholtz, “GTDB: An ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy,” *Nucleic Acids Research*, vol. 50, no. D1, pp. D199–D207, 2022. doi: [10.1093/nar/gkab776](https://doi.org/10.1093/nar/gkab776)
- [39] J. Pfeiffer, A. Kamath, A. Rücklé, K. Cho, and I. Gurevych, “AdapterFusion: Non-destructive task composition for transfer learning,” *arXiv preprint arXiv:2005.00247*, 2021.
- [40] Human Microbiome Project Consortium, “Structure, function and diversity of the healthy human microbiome,” *Nature*, vol. 486, no. 7402, pp. 207–214, 2012. doi: [10.1038/nature11234](https://doi.org/10.1038/nature11234)





# Appendix A

## Experiment Configuration Details



This appendix provides the full configuration files used for the key experiments described in Chapter 4.

### A.1 v8 Genus Configuration (5M Balanced)

```
1 model:
2   backbone: InstaDeepAI/nucleotide-transformer-v2-500m-multi-
3     species
4   head_type: attention_pool
5   num_query_tokens: 4
6   hidden_dim: 512
7   dropout: 0.15
8
9 lora:
10   r: 16
11   alpha: 32
12   dropout: 0.05
13   target_modules: [query, key, value]
14
15 data:
16   fasta_path: /path/to/balanced_5M/reads.fa
17   labels_path: /path/to/balanced_5M/labels.tsv
18   classification_level: genus
19   max_length: 32
20   val_ratio: 0.1
21
22 training:
23   batch_size: 32
```



```
23 gradient_accumulation_steps: 2
24 num_epochs: 20
25 phase1_epochs: 5
26 backbone_lr: 3.0e-5
27 head_lr: 5.0e-4
28 weight_decay: 0.02
29 warmup_ratio: 0.1
30 early_stopping_patience: 7
31 rc_augmentation: true
```

Listing A.1: YAML configuration for v8 genus experiment.

## A.2 v9 Genus Configuration (50M Balanced)

```
1 model:
2   backbone: InstaDeepAI/nucleotide-transformer-v2-500m-multi-
   species
3   head_type: attention_pool
4   num_query_tokens: 4
5   hidden_dim: 512
6   dropout: 0.15
7
8   lora:
9     r: 16
10    alpha: 32
11    dropout: 0.05
12    target_modules: [query, key, value]
13
14 data:
15   fasta_path: /path/to/balanced_50M/reads.fa
16   labels_path: /path/to/balanced_50M/labels.tsv
17   classification_level: genus
18   max_length: 32
```



```
19   val_ratio: 0.1
20
21 training:
22   batch_size: 256
23   gradient_accumulation_steps: 1
24   num_epochs: 30
25   phase1_epochs: 0
26   backbone_lr: 3.0e-5
27   head_lr: 5.0e-4
28   weight_decay: 0.02
29   warmup_ratio: 0.1
30   early_stopping_patience: 7
31   rc_augmentation: true
```

Listing A.2: YAML configuration for v9 genus experiment.

### A.3 SLURM Job Script Example

```
1 #!/bin/bash
2 #SBATCH --job-name=gfm-genus-v9
3 #SBATCH --partition=gpu
4 #SBATCH --gres=gpu:1
5 #SBATCH --cpus-per-task=4
6 #SBATCH --mem=128G
7 #SBATCH --time=48:00:00
8 #SBATCH --output=slurm-%j.out
9
10 module load cuda/12.1
11 source activate gfm
12
13 python scripts/train.py \
14   --config configs/nt_token_genus_v9_50M.yaml \
15   --init_from results/nt_token_genus_lora_v8_5M/best.pt
```

16  
17  
18  
19  
20  
21  
22  
23  
24

```
python scripts/evaluate.py \  
    --config configs/nt_token_genus_v9_50M.yaml \  
    --checkpoint results/nt_token_genus_lora_v9_50M/best.pt  
  
python scripts/evaluate.py \  
    --config configs/nt_token_genus_v9_50M.yaml \  
    --checkpoint results/nt_token_genus_lora_v9_50M/best.pt \  
    --rc_tta
```



Listing A.3: SLURM script for training on TWCC cluster.